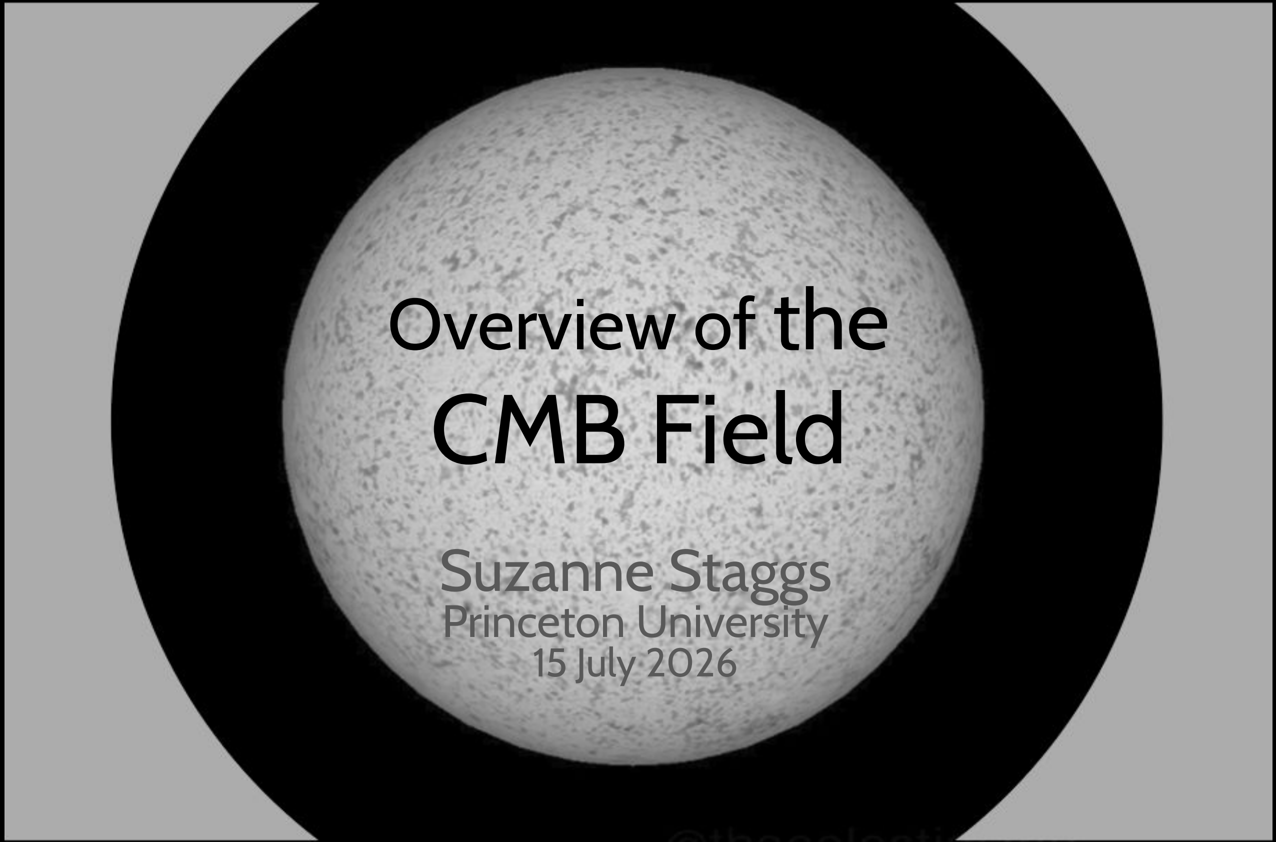


Overview of the CMB Field

Suzanne Staggs
Princeton University

15 July 2026

2026 Flatiron Institute CMB Summer School

PROUST

Remembrance of Things Past

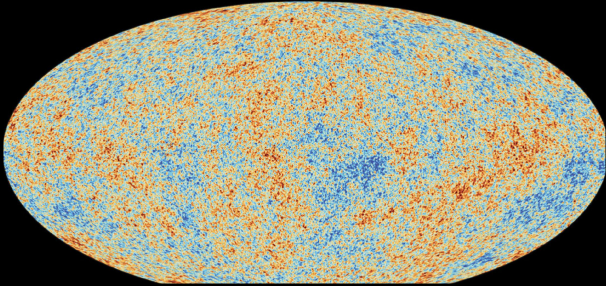
Volume Three

THE PRIMEVAL FIREBALL

The earth is bathed in radio waves that appear to have originated at the time of the primordial "big bang." This radiation provides the cosmologist with a rare new clue to the nature of the universe

by P. J. E. Peebles and David T. Wilkinson

Scientific American, June 1967



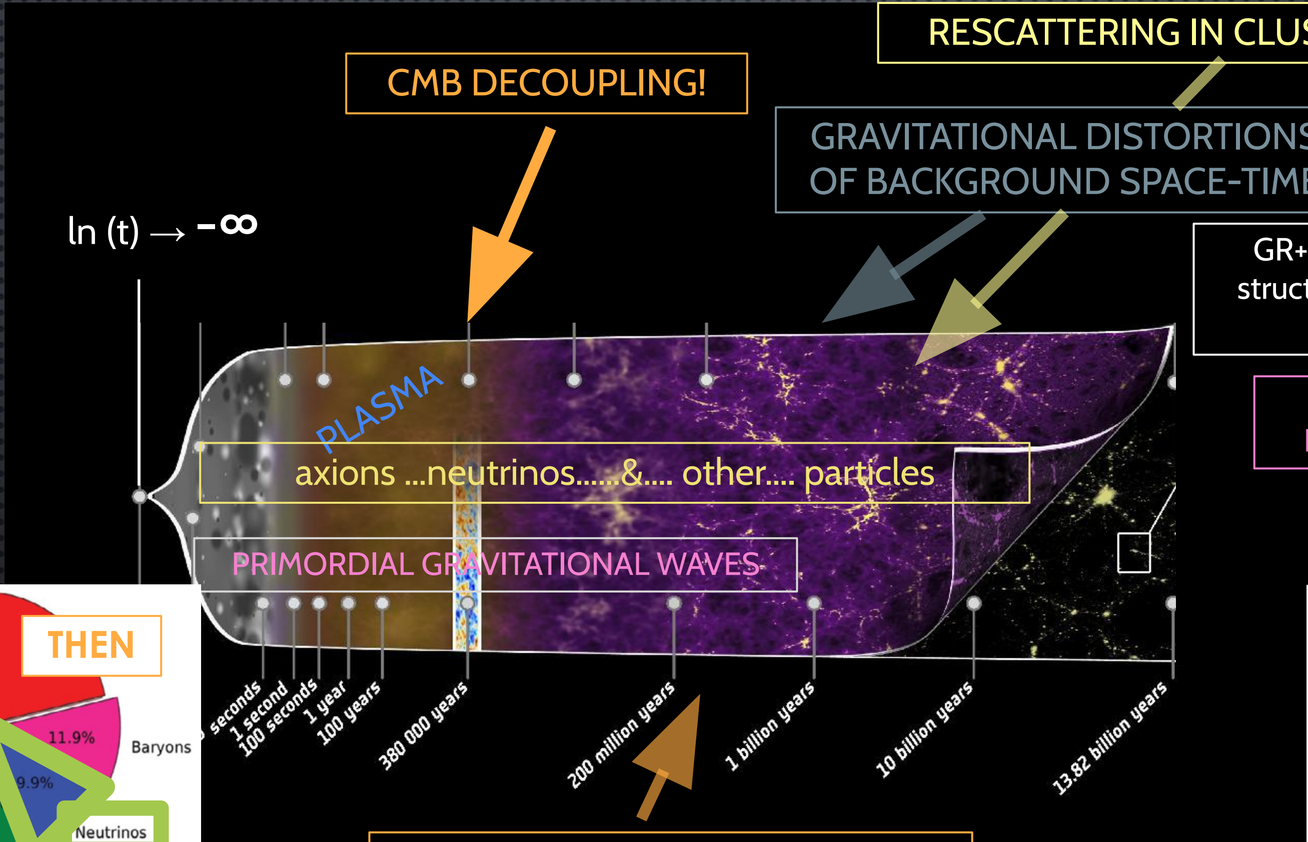
2013: Courtesy Planck Science Team

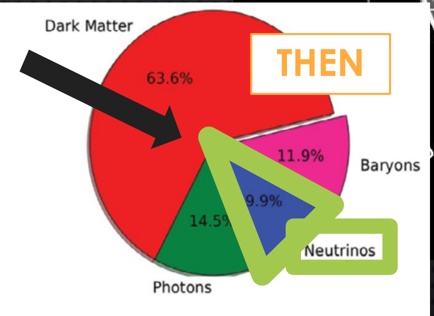
OUTLINE:

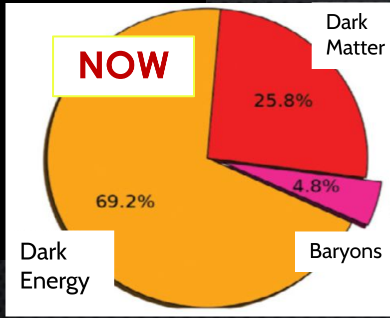
1) **CMB as treasure trove / layer cake / archaeological strata**
2) Upcoming new CMB data / instruments in brief
3) Polarization, Parity, Fourier Space, Real Space, Q, U, E, B
4) Deeper dive on CMB instruments

SOURCES OF EVIDENCE IN THE CMB

Base image courtesy of Planck science team







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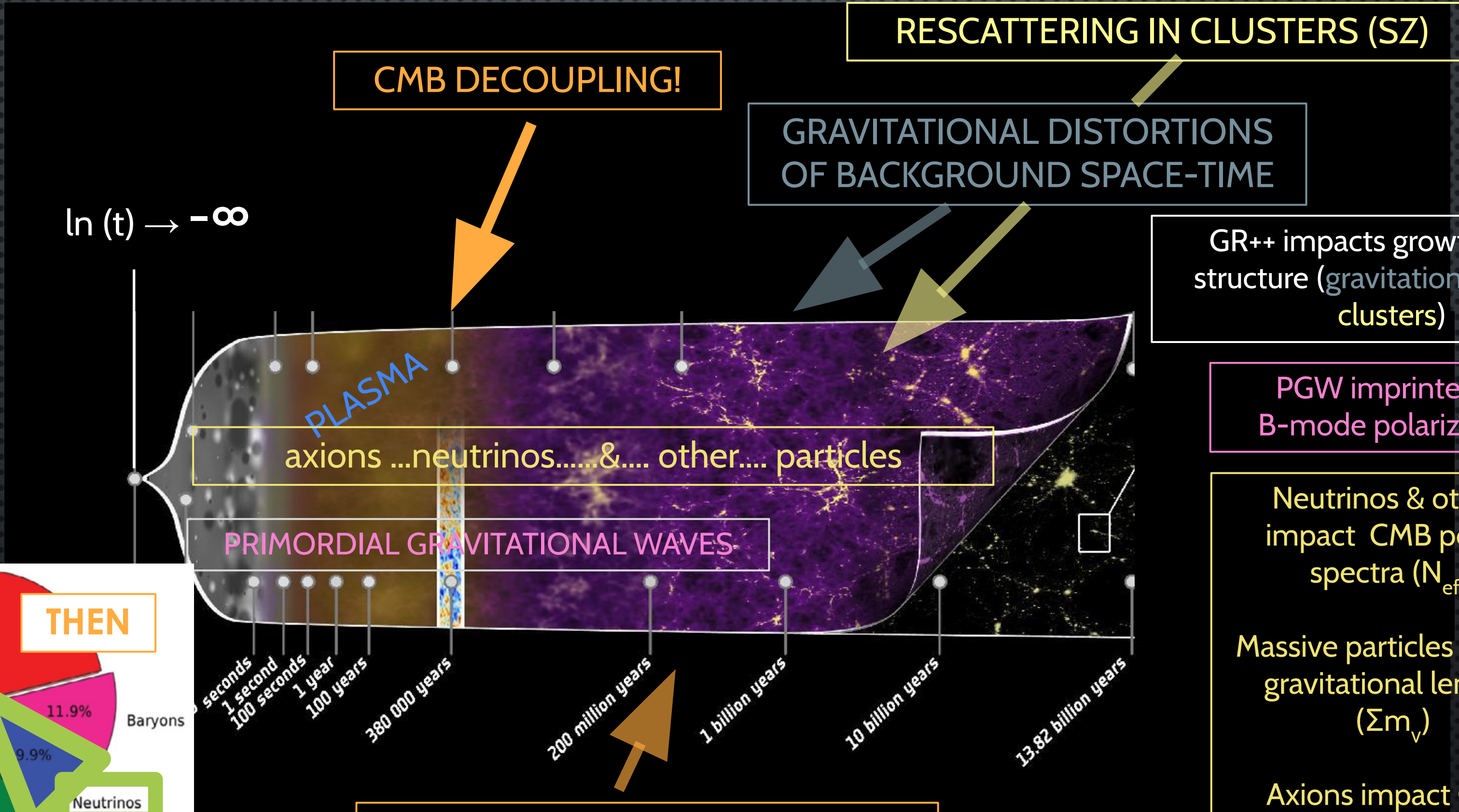


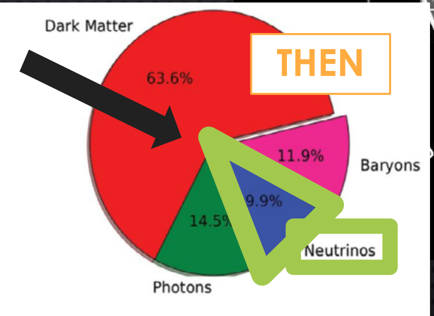
GR++ impacts growth rate of structure (gravitational lensing, clusters)

PGW imprinted in B-mode polarization

Neutrinos & others impact CMB power spectra (N_{eff})

Massive particles impact gravitational lensing (Σm_ν)

Axions impact CMB power spectra and polarization



Next Up:
Overview of CMB Impact on Cosmology

Overview of CMB Impact on Cosmology

via Screen Shots from Mat Madhavacheril's talk yesterday!

The WMAP+Planck Legacy -> current generation CMB

- Precision concordance ΛCDM model
- Init. Conditions: Gaussian, adiabatic, slightly red near scale-invariant power
- Dark matter needed to explain acoustic oscillations
- Acceleration due to dark energy (indep. evidence from CMB)
- Three relativistic/light neutrinos, $\mathbf{N_{eff}}$~3

- Explore space of models for $\mathbf{H_0}$ tension
- New light particles $\mathbf{\Delta N_{eff}}$
- Dark matter interactions
- Inflationary gravitational waves, **r** with B-modes
- Reconstructions and cross-correlations of mass (lensing), gas pressure (tSZ), velocities and gas density (kSZ)
  - Measure neutrino mass $\mathbf{\Sigma m_\nu}$
  - Precision tests of non-Gaussianity from multi-field inflation $\mathbf{f_{NL}}$
  - Measure baryonic feedback, constrain galaxy formation

OUTLINE:

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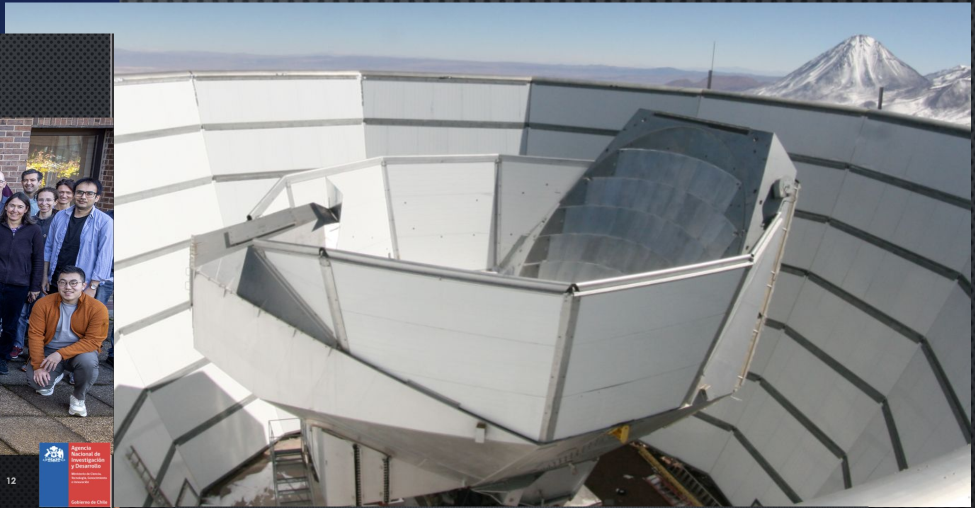
Staggs Bona Fides & Biases (1)



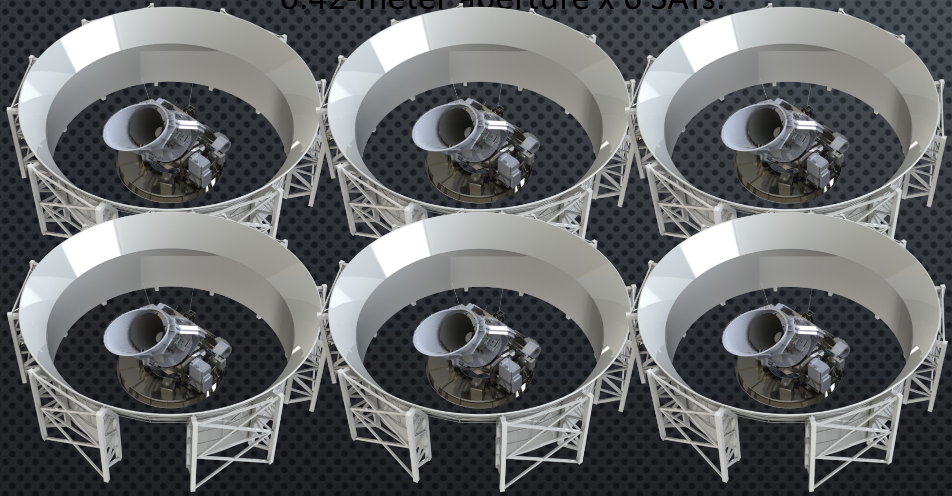
ATACAMA COSMOLOGY TELESCOPE 2007–2022

6 m off-axis telescope (arcminute resolution) formerly @ 5200 m (high) in a desert (dry) at latitude -23° (north-ish)

Staggs Bona Fides & Biases (2)

Small Aperture Telescopes (SATs)

- 0.42-meter aperture x 6 SATs.



Large Aperture Telescope (LAT)

- 6-meter aperture (huge focal plane)

SIMONS OBSERVATORY

@ 5200 m (high) in a desert (dry) at latitude -23° (north-ish)

Staggs Bona Fides & Biases (2)

> 450 collaboration members (300 with PhDs)

SIMONS OBSERVATORY

@ 5200 m (high) in a desert (dry) at latitude -23° (north-ish)

TAXONOMY OF UPCOMING CMB DATA

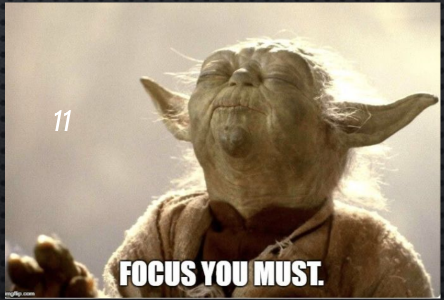
High resolution maps

- Arcminute resolution
- Deep and wide
- Multiple frequencies spanning 20-300 GHz
- Temperature & (linear) polarization
- High cadence time resolution capabilities

Fodder for myriad studies

Polarization data

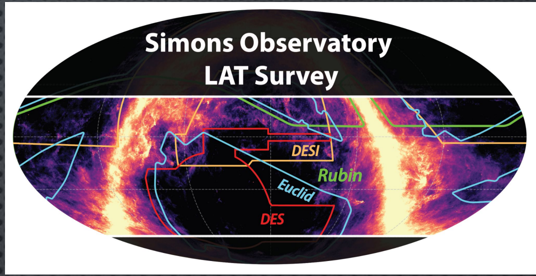
- Very deep maps
- Degree scale resolution
- High fidelity recovery of low ells
- Multiple frequencies spanning 20-300 GHz



PAROCHIAL EXAMPLE OF SO PLANNED SURVEYS

High resolution maps

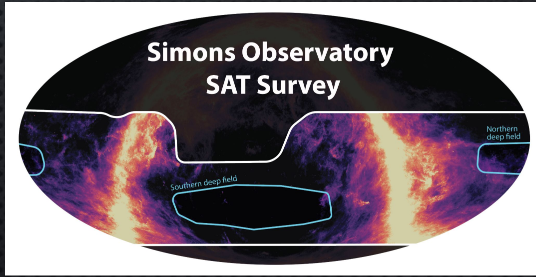
- Arcminute resolution
- Deep and wide
- Multiple frequencies spanning 20-300 GHz
- Temperature & (linear) polarization
- High cadence time resolution capabilities



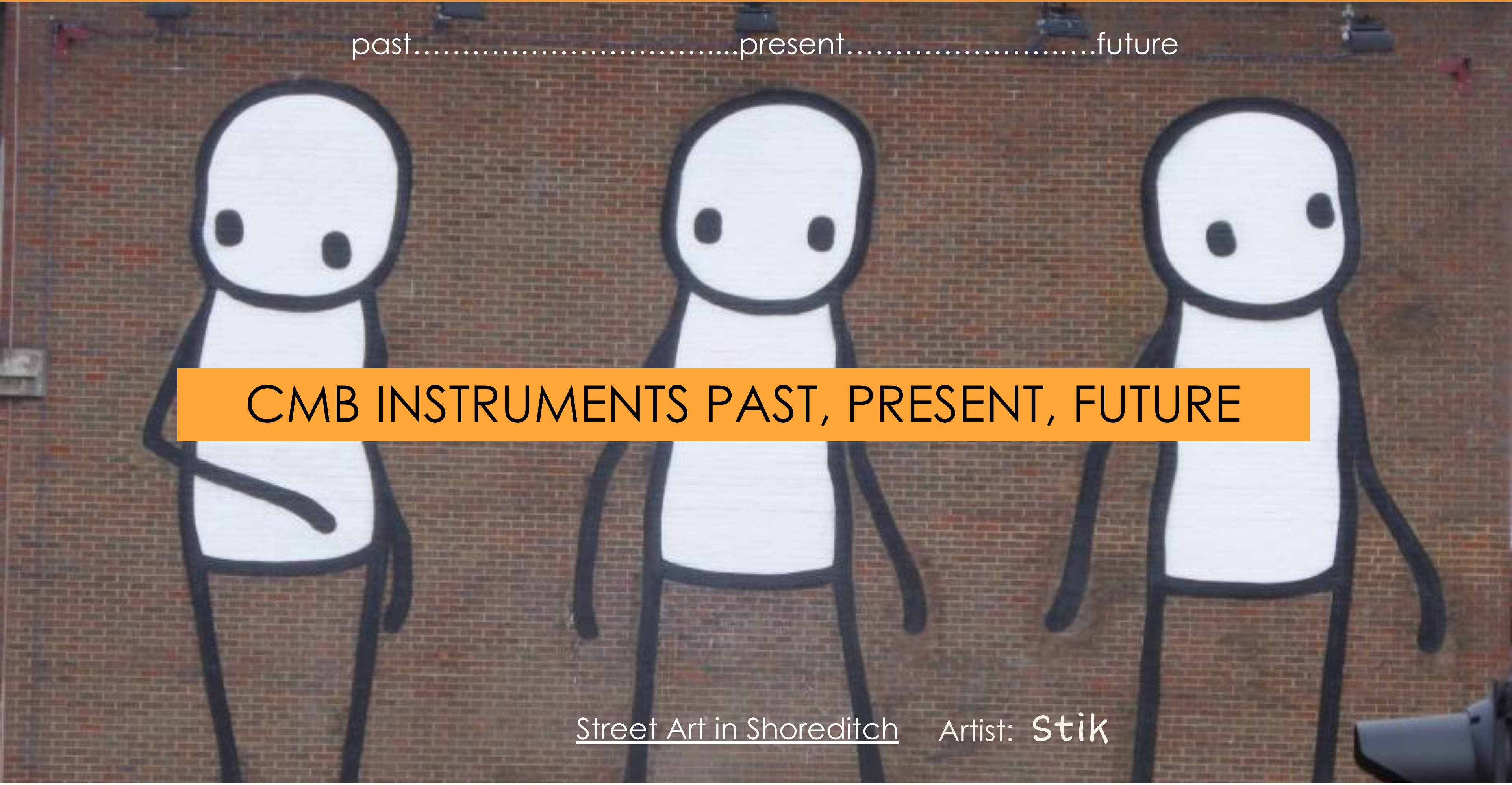
and also
SPO (BICEP & SPT-3G+)
ALI-CPT, TAURUS, QUBIC +

Polarization data

- Very deep maps
- Degree scale resolution
- High fidelity recovery of low ells
- Multiple frequencies spanning 20-300 GHz



past...................................present...........................future



CMB INSTRUMENTS PAST, PRESENT, FUTURE

Street Art in Shoreditch Artist: Stik

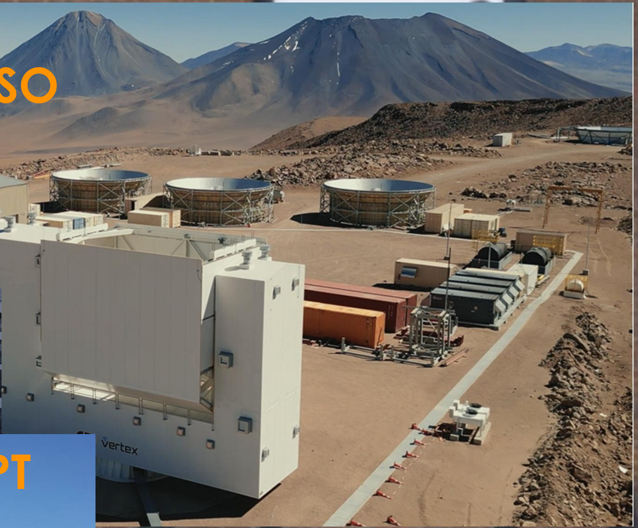
past..................................present.............................future

RELIKT-1 (1983-84)
COBE (1989-93)
COBRA (1990)
WMAP (2001-10)
Planck (2009-13)

LiteBIRD (2032+)
others proposed!

ARCADE ARCHEOPS
BOOMERANG EBEX
FIRS MAXIMA MAXIPOL
MSAM PIPER QMAP
QUAD TOPHAT

ABS ACBAR ACME
AMiBA APEX BAM
BEAST BICEP1,2
CAPMAP CBI CG
COMPASS
COSMOSOMAS DASI
KECK ARRAY KUPID
MAT MINT PIQUE
POLAR POLARBEAR
PYTHON QUAD QUIET
QUIJOTE Saskatoon
SZA SuZIE Tenerife VSA



Tenerife

TAURUS

ALI-CPT

QUBIC

OUTLINE:

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STOKES PARAMETERS

Define local x,y system

$$I = \langle E_x^2 + E_y^2 \rangle$$

$$Q = \langle E_x^2 - E_y^2 \rangle$$

$$U = E_x E_y \cos\phi$$

$$V = E_x E_y \sin\phi$$

Q & U are LOCAL; they can be defined at a point on the sky (as can I & V).

STOKES PARAMETERS vs. E & B MODES (FLAT SKY)

Define local x,y system

$$I = \langle E_x^2 + E_y^2 \rangle$$

$$Q = \langle E_x^2 - E_y^2 \rangle$$

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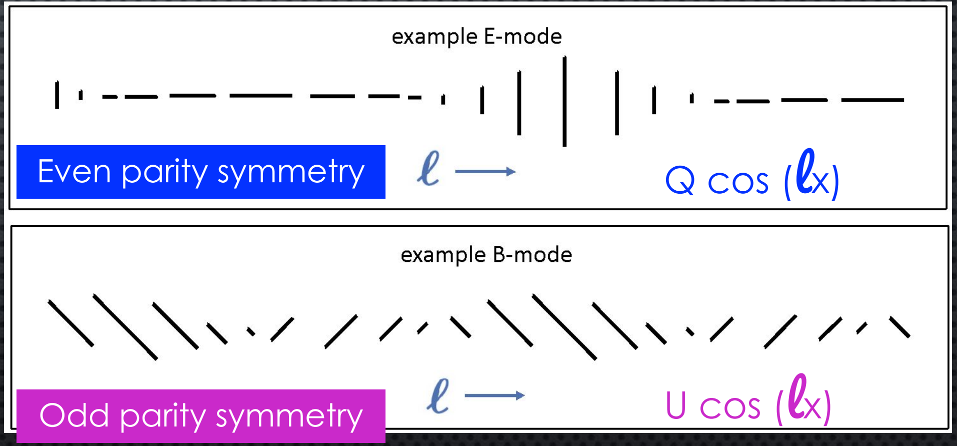


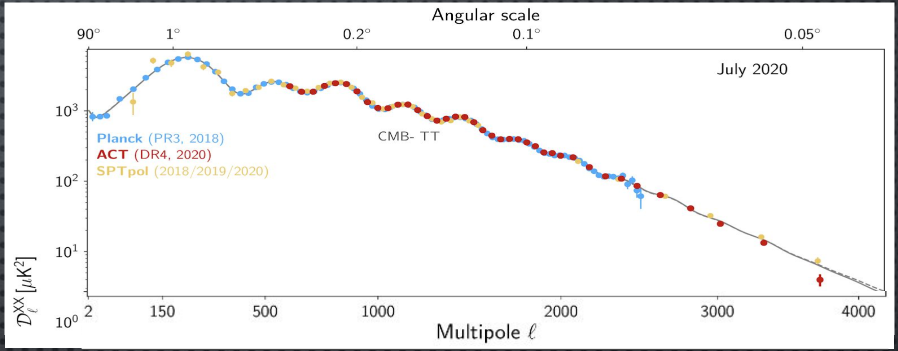
E & B are GLOBAL rather than local since they are fundamentally defined in the harmonic space.

REAL SPACE, HARMONIC SPACE: ACOUSTIC OSCILLATIONS

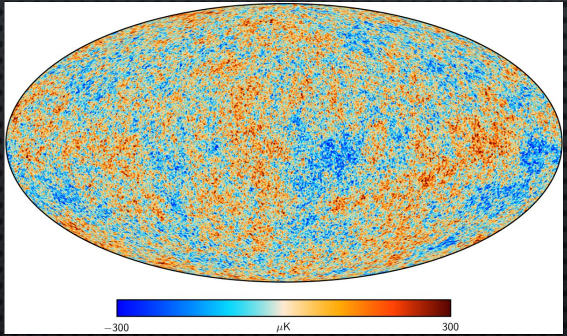
Wiener-Khinchin Theorem

autocorrelation $$r_{xx}(\tau) = \int_{-\infty}^{\infty} S(f) e^{2\pi i \tau f}\, df.$$

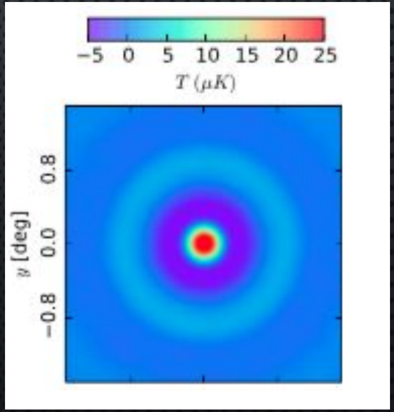
power spectrum $$S(f) = \int_{-\infty}^{\infty} r_{xx}(\tau) e^{-2\pi i f \tau}\, d\tau.$$



Harmonic transform



Stack around hot spots



ACT DR3 IN REAL SPACE

LOUIS ET AL 2016: 1610.02360

CMB POLARIZATION SCHEMATICALLY

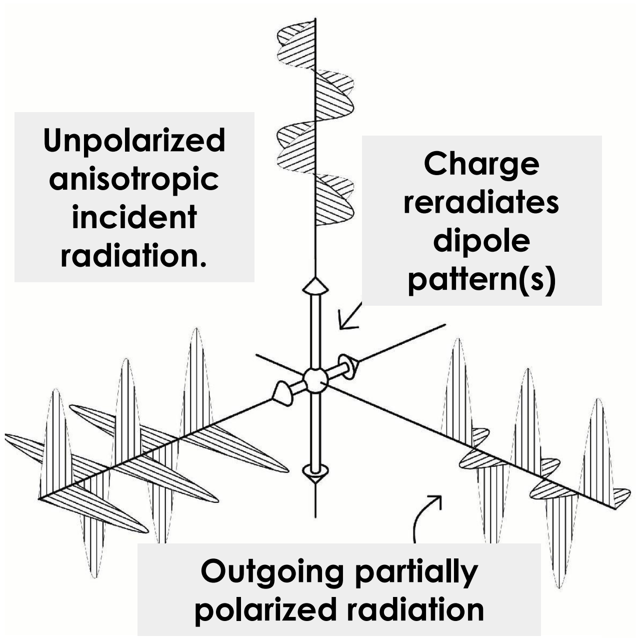


Figure courtesy of M. Hedman

The CMB is slightly polarized by Thomson scattering whenever there is any local quadrupolar anisotropy in the distribution of the scattering electron population.

Acoustic oscillations create local quadrupoles

ACOUSTIC OSCILLATIONS & POLARIZATION PATTERNS

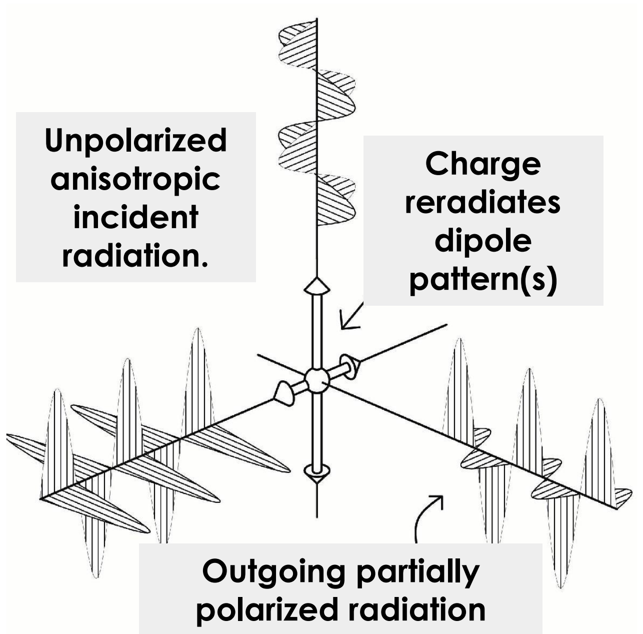
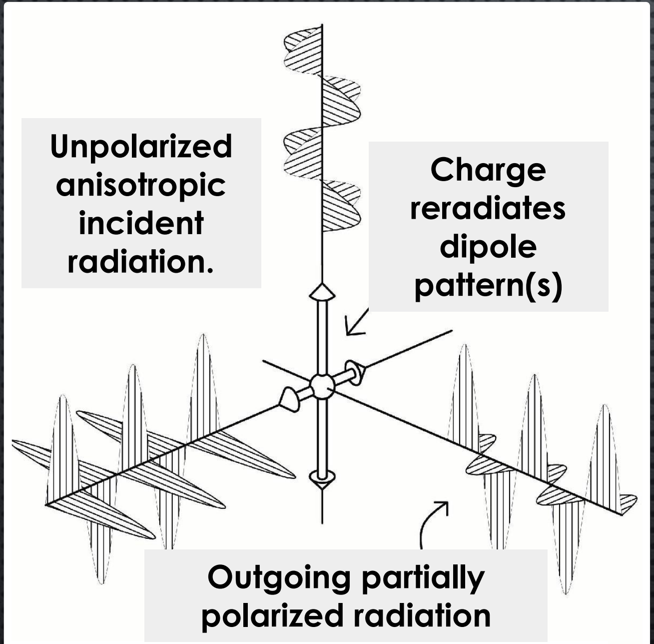


Figure courtesy of M. Hedman

Consider a mass overdensity as plasma begins flowing in

ACOUSTIC OSCILLATIONS & POLARIZATION PATTERNS



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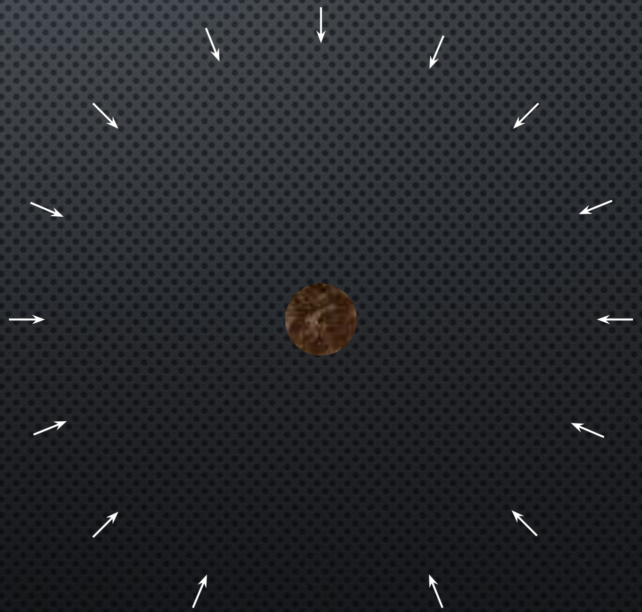

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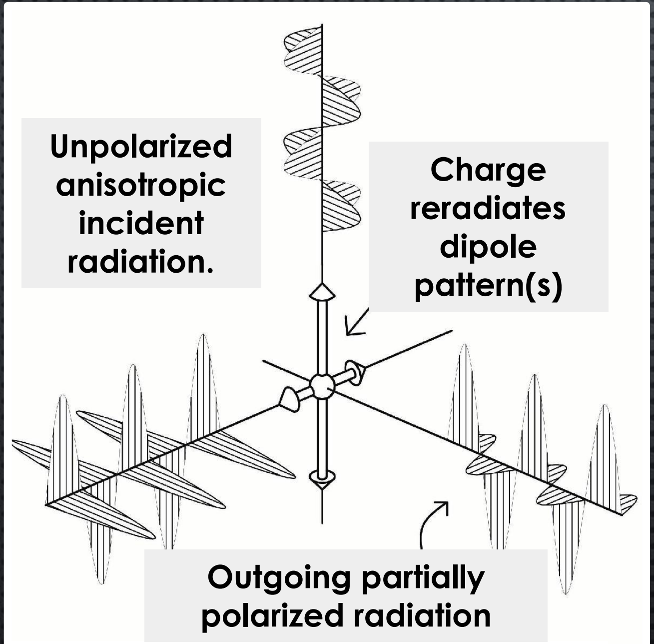

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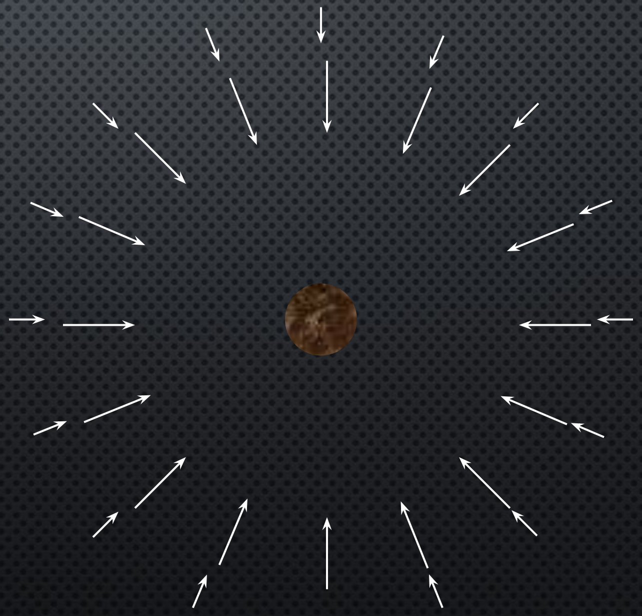

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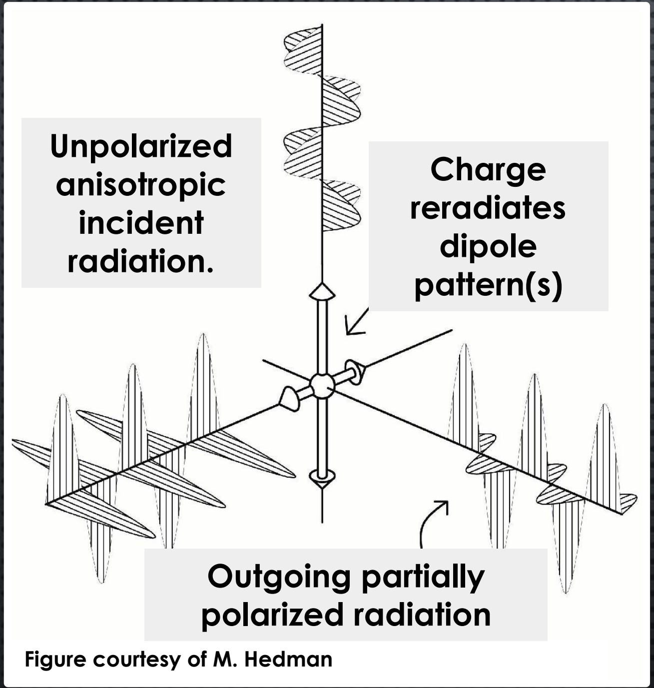

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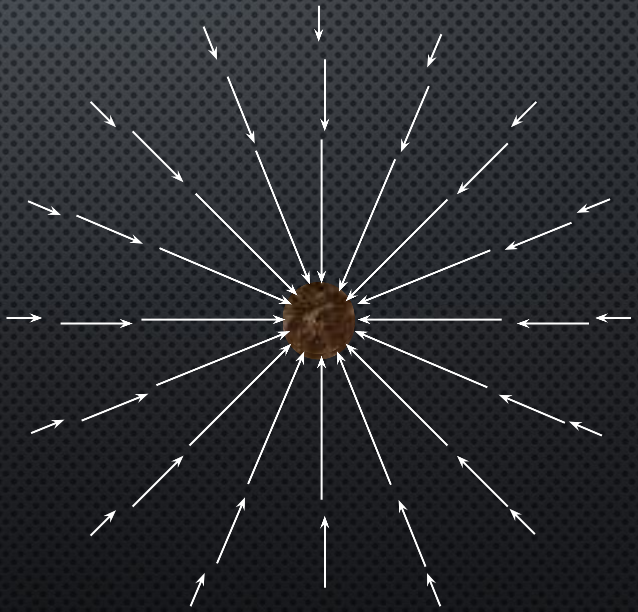

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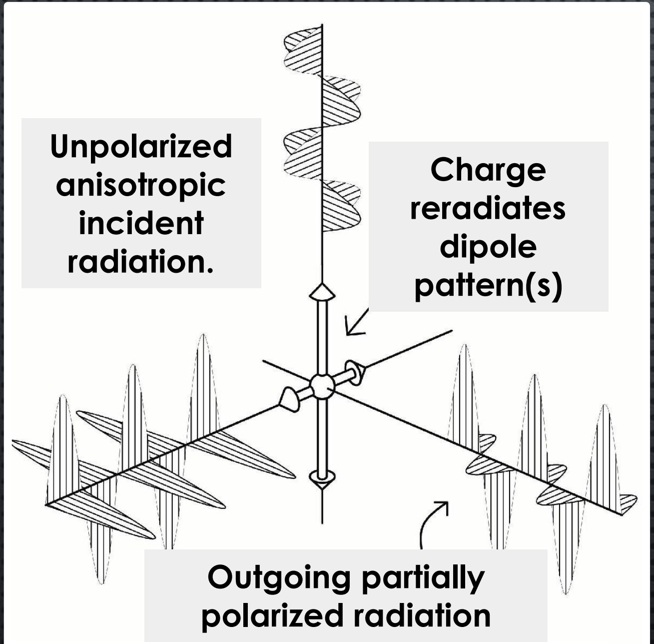

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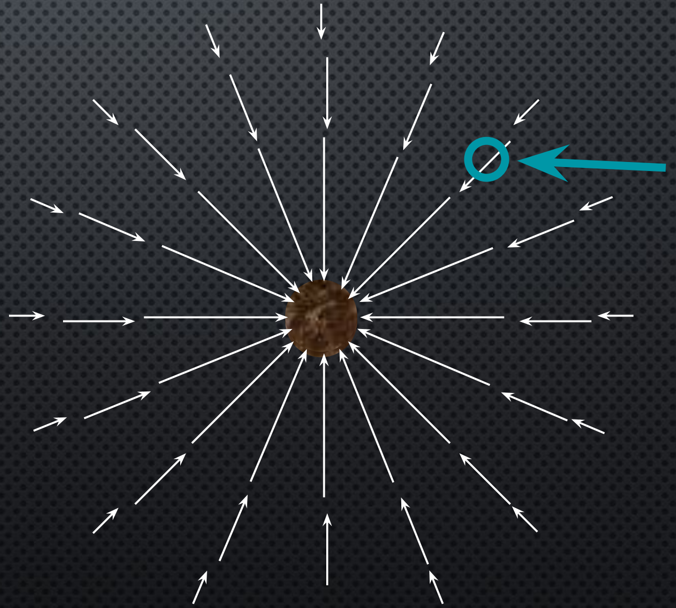
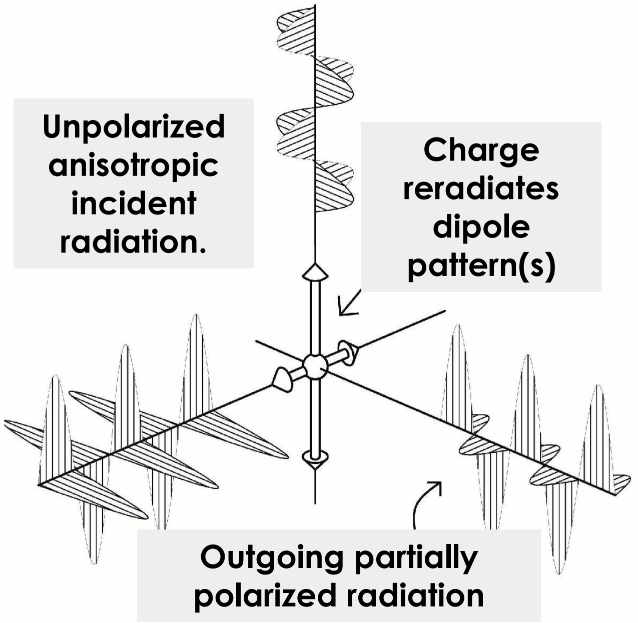


Consider an electron in this part of the flowing plasma

ACOUSTIC OSCILLATIONS & POLARIZATION PATTERNS



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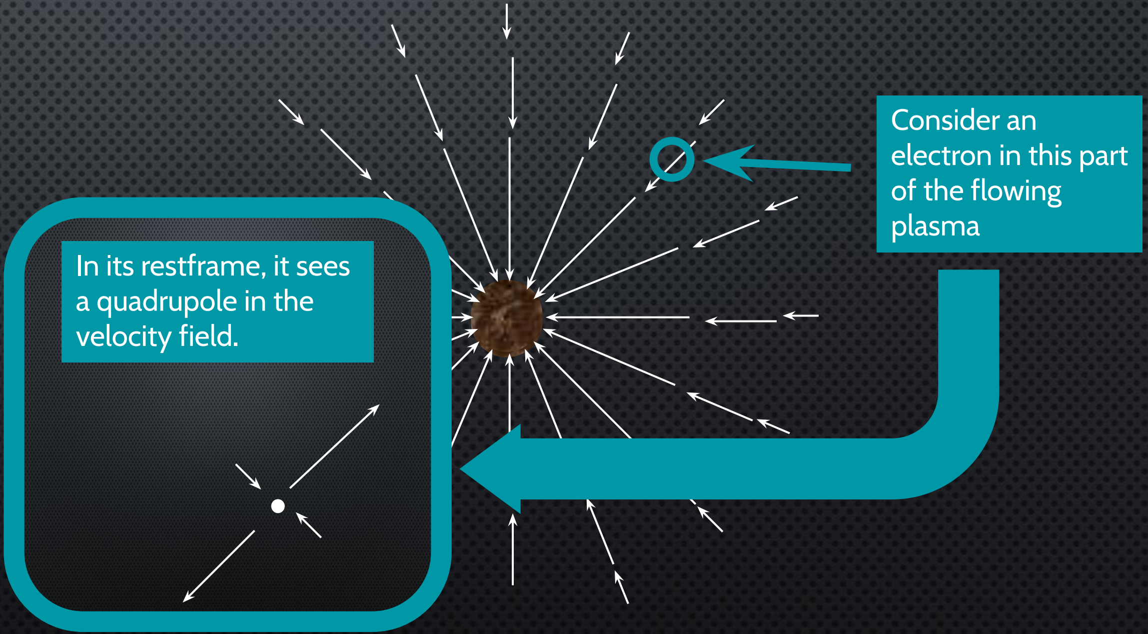

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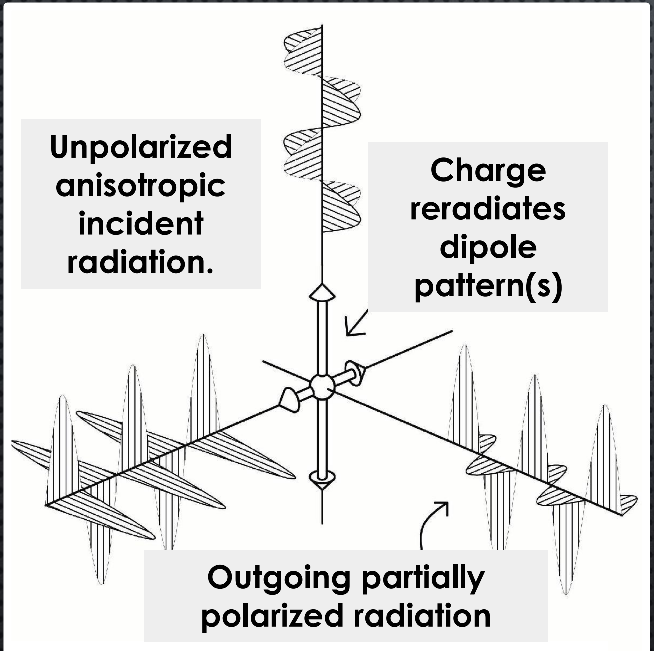
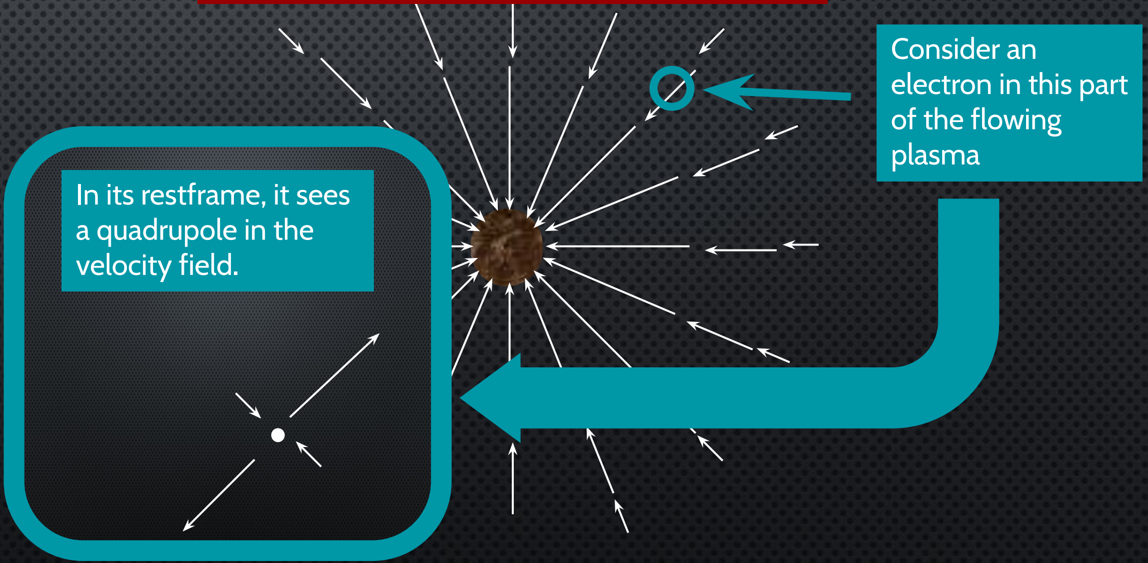


Figure courtesy of M. Hedman

Acoustic oscillations create local quadrupoles (Doppler effect)

Consider a mass overdensity as plasma begins flowing in

Doppler shifts provide the requisite intensity quadrupole!



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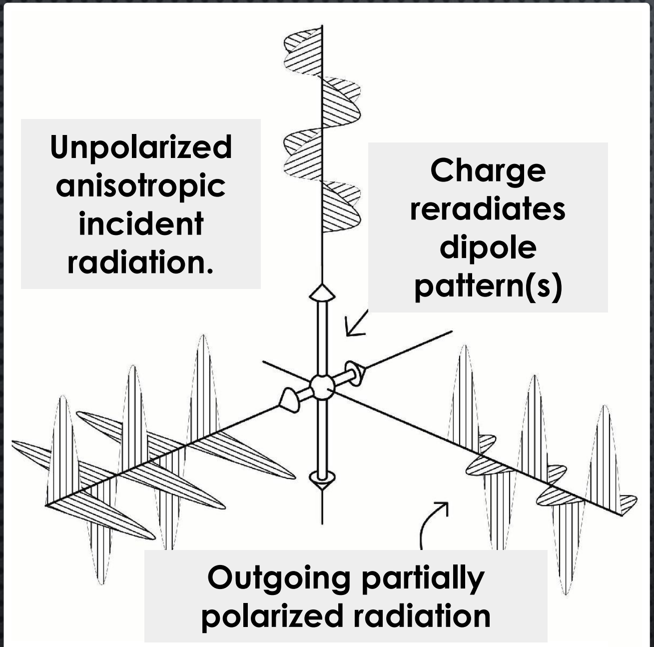

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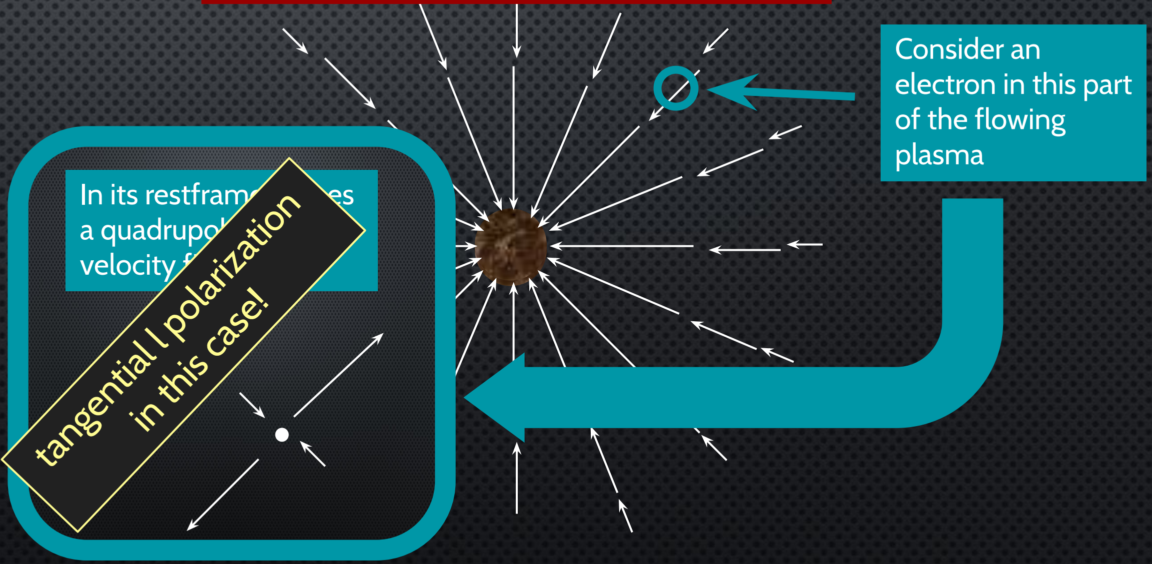


CMB POLARIZATION SCENARIO

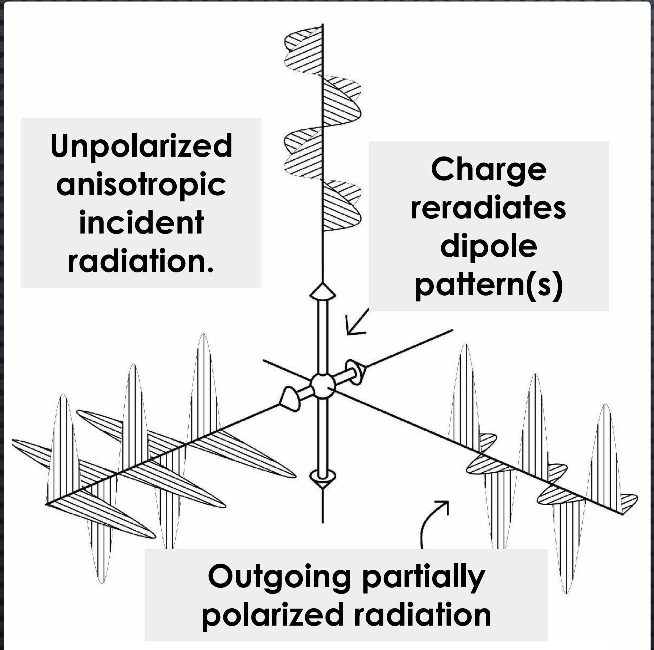
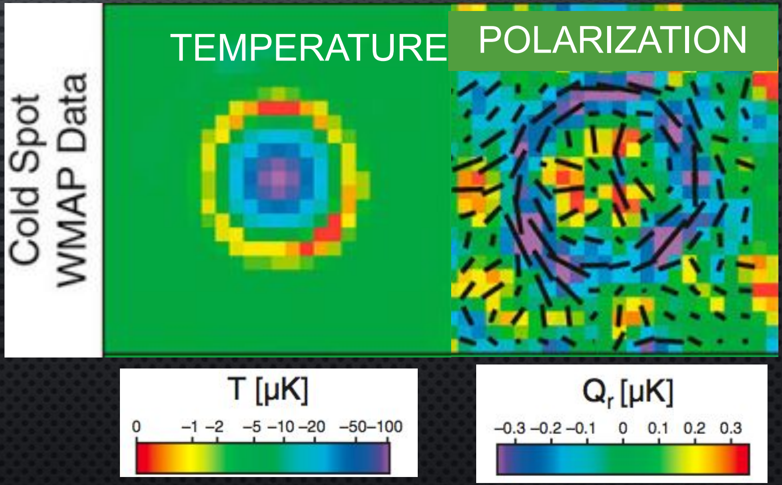


Figure courtesy of M. Hedman

Acoustic oscillations create local quadrupoles (Doppler effect)

WMAP7 stacking on cold spots
Komatsu et al, 2011.



CMB POLARIZATION SCENARIO

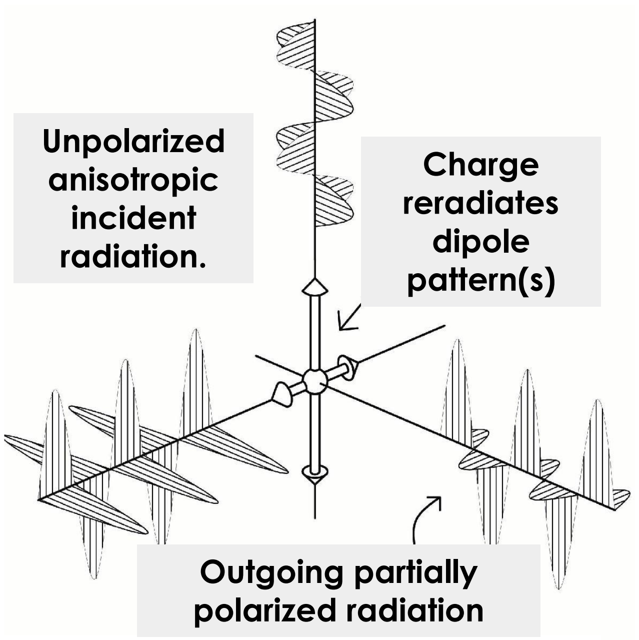
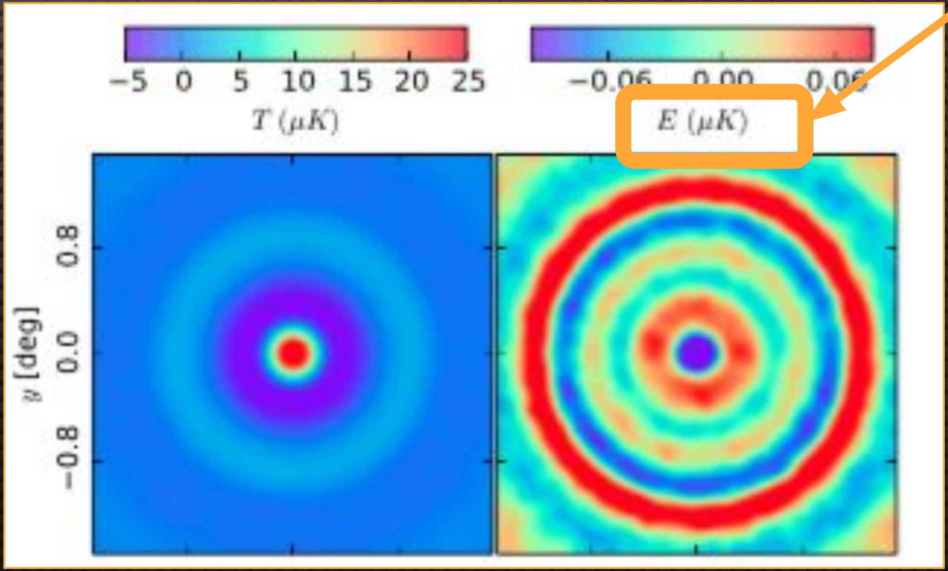


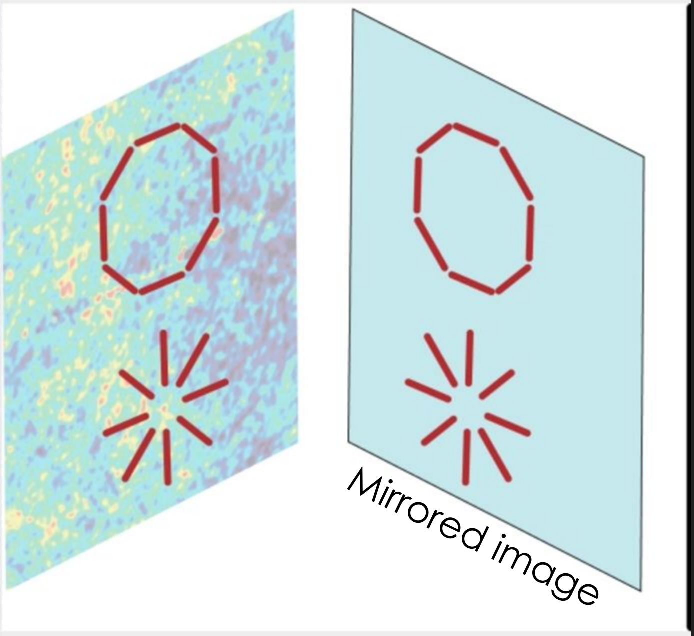
Figure courtesy of M. Hedman

Acoustic oscillations create local quadrupoles (Doppler effect)

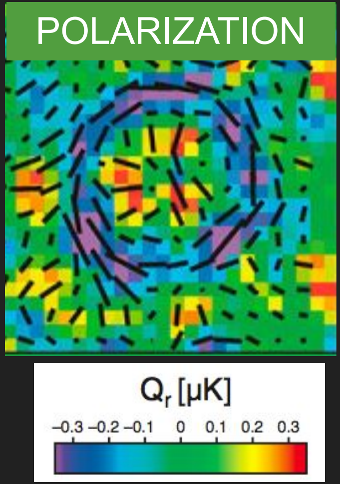
and later:
ACT DR3 IN REAL SPACE
LOUIS ET AL 2016: 1610.02360



ACOUSTIC OSCILLATIONS PRODUCE E-MODES (WITH EVEN PARITY SYMMETRY)

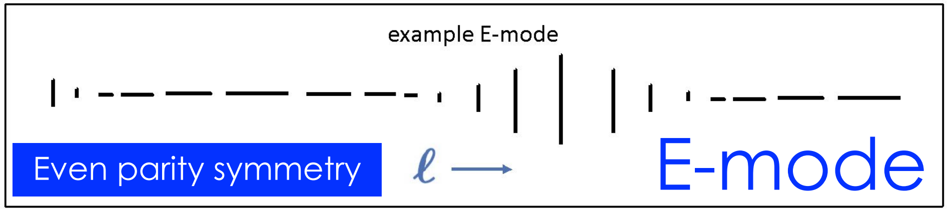


Radial & tangential polarization patterns around cold/ hot spots



Mirror artwork from **Shane L. Larson**: https://writescience.wordpress.com/tag/gravitational-waves/

ACOUSTIC OSCILLATIONS PRODUCE E-MODES (WITH EVEN PARITY SYMMETRY)

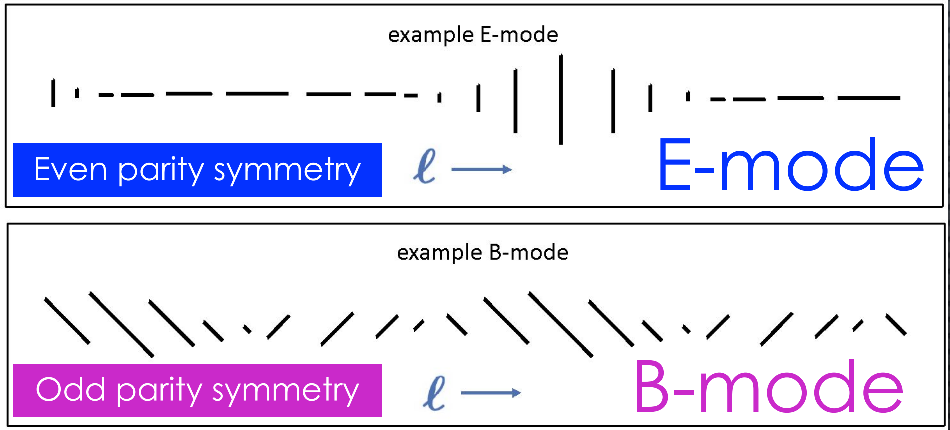


Inflation (or other P(k) progenitor) could cause primordial gravitational waves (PGW).

GRAVITATIONAL WAVES (TENSOR MODES) PRODUCE E AND B

ACOUSTIC OSCILLATIONS (SCALAR MODES) PRODUCE ONLY E-MODES!

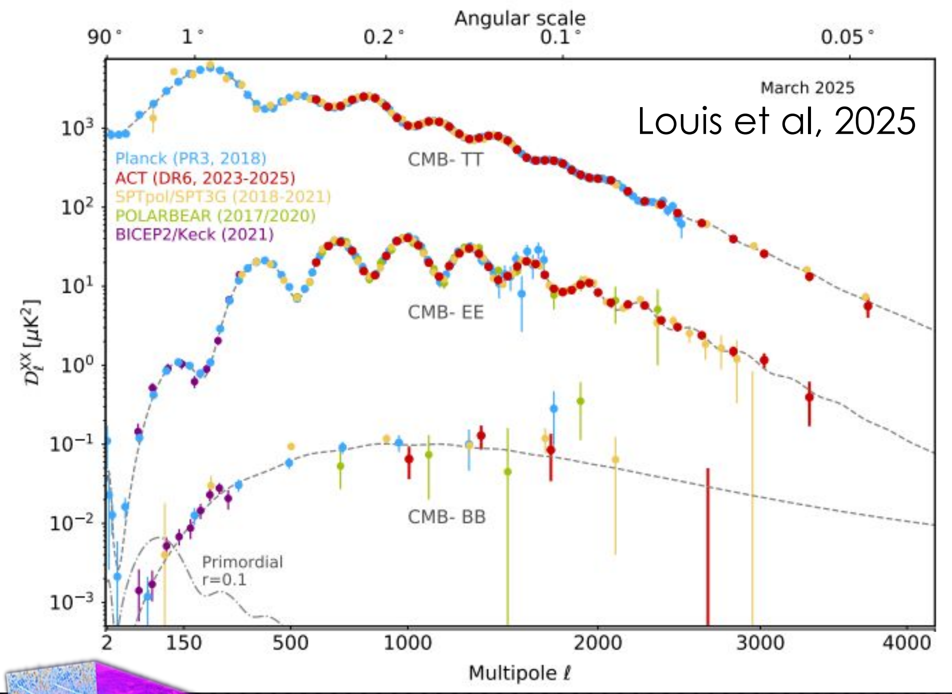
Kamionkowski, Kosowsky & Stebbins; 1997; Seljak & Zaldarriaga 1997

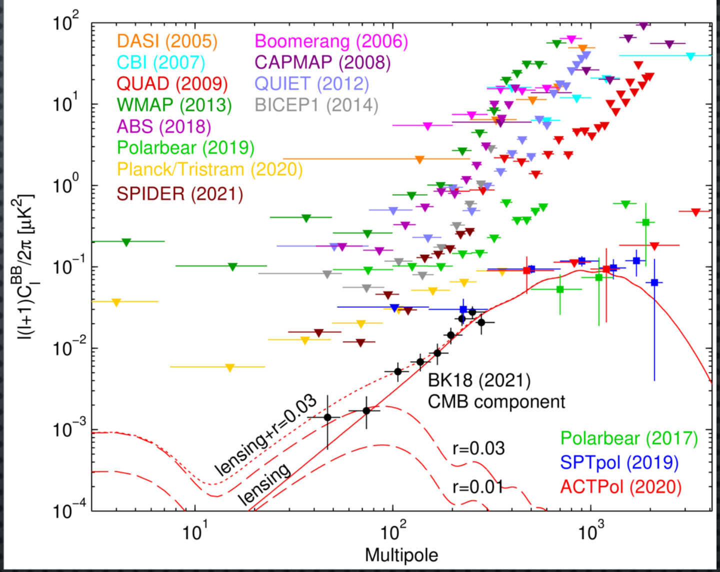


Inflation (or other P(k) progenitor) could cause primordial gravitational waves (PGW).

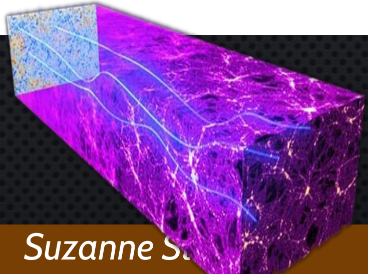
GRAVITATIONAL WAVES (TENSOR MODES) PRODUCE E AND B

MORE ON THE POLARIZATION DATA:





BICEP/Keck Collaboration, PRL 127, 151301,2021

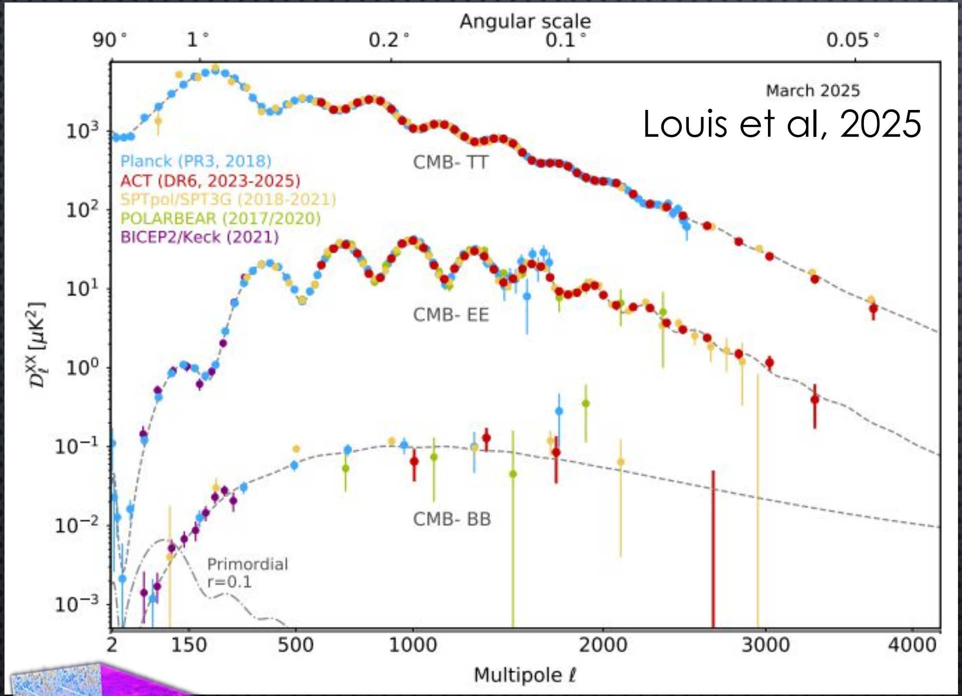


Gravitational lensing
re-arranges spatial patterns,
converting E $\rightarrow$B

Current limit on tensor-to-scalar power
spectrum ration at k = 0.05 Mpc^{-1}:

r< 0.036, 95% CL.

A FEW MORE THINGS ABOUT CMB POLARIZATION DATA:



- Primordial gravitational wave effects at low ell → small aperture telescopes
- EE, TE provide new draws of the underlying Gaussian processes →**high ell is rich!**
- Smaller foregrounds so cleaner→ **high ell is rich!**
- Eventually **lensing** signal will be better than from TT → **high ell is rich!**

Gravitational lensing arises from dozens of arcminute deflections of geodesics

One person's fish is another person's poison.

CMB POLARIZATION: E & B on Celestial Sphere

TEMPERATURE IS A SCALAR

$$\delta T(\theta,\varphi) = \sum_{l,m} a_{lm} Y_{lm}(\theta,\varphi)$$

Nice form!

LINEAR POLARIZATION IS A TENSOR :Requires a magnitude **and** a direction.

Spin-weighted harmonics

$$(Q+iU)(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{(\pm 2)} \left[{}_{\pm 2}Y_{\ell m}(\hat{n})\right]$$

$$a_{\ell m}^{E} \equiv -\frac{1}{2}\left(a_{\ell m}^{(2)} + a_{\ell m}^{(-2)}\right), \qquad a_{\ell m}^{B} \equiv -\frac{1}{2i}\left(a_{\ell m}^{(2)} - a_{\ell m}^{(-2)}\right)$$

E & B: SCALARS FOR POLARIZATION)

Nice form!

$$E(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{E}\, Y_{\ell m}(\hat{n}), \qquad B(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{B}\, Y_{\ell m}(\hat{n})$$

Seljak & Zaldarriaga 1997; Kamionkowski, Kosowsky, Stebbins 1997 (E→G, B→C)

CMB POLARIZATION: E & B cf Q & U

Full relationships: Seljak & Zaldarriaga 1997 (and Kamionkowski, Kosowsky, Stebbins 1997)

$$(Q + iU)(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{(\pm 2)} \left[{}_{\pm 2}Y_{\ell m}(\hat{n})\right]$$

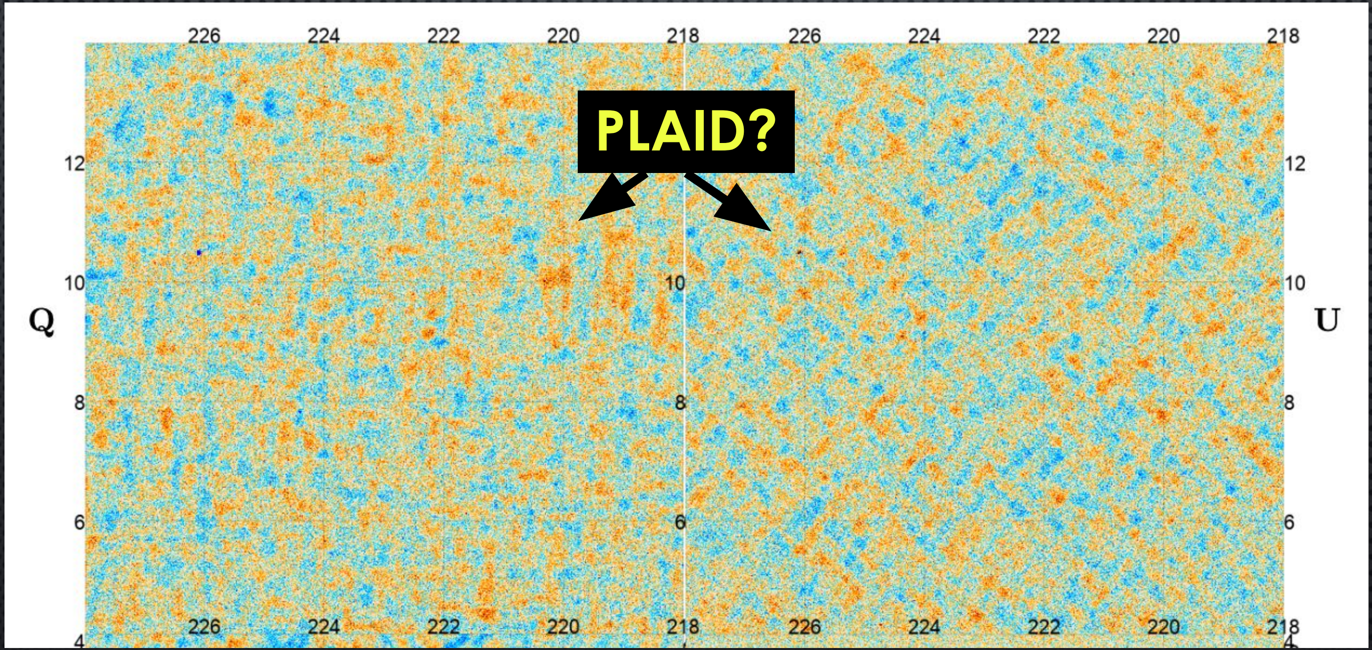
$$a_{\ell m}^{E} \equiv -\frac{1}{2}\left(a_{\ell m}^{(2)} + a_{\ell m}^{(-2)}\right), \qquad a_{\ell m}^{B} \equiv -\frac{1}{2i}\left(a_{\ell m}^{(2)} - a_{\ell m}^{(-2)}\right)$$

$$E(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{E}\, Y_{\ell m}(\hat{n}), \qquad B(\hat{n}) = \sum_{\ell,m} a_{\ell m}^{B}\, Y_{\ell m}(\hat{n})$$

Flat sky approximations: Zaldarriaga, 2001, PRD 64

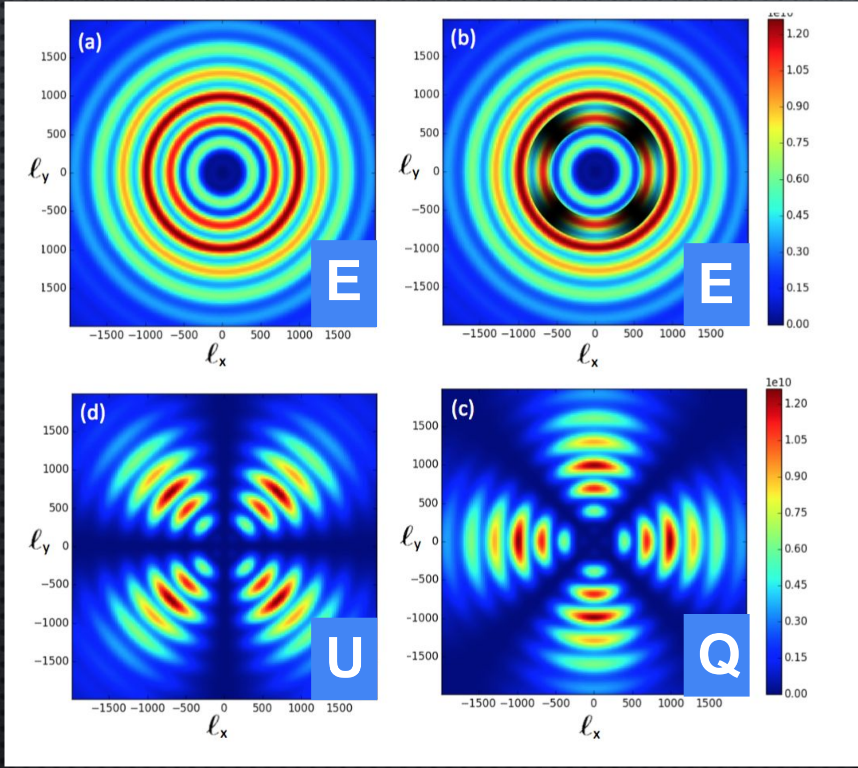
$$Q(\boldsymbol{\ell}) = E(\boldsymbol{\ell})\cos(2\phi_\ell) - B(\boldsymbol{\ell})\sin(2\phi_\ell),$$
$$U(\boldsymbol{\ell}) = E(\boldsymbol{\ell})\sin(2\phi_\ell) + B(\boldsymbol{\ell})\cos(2\phi_\ell).$$

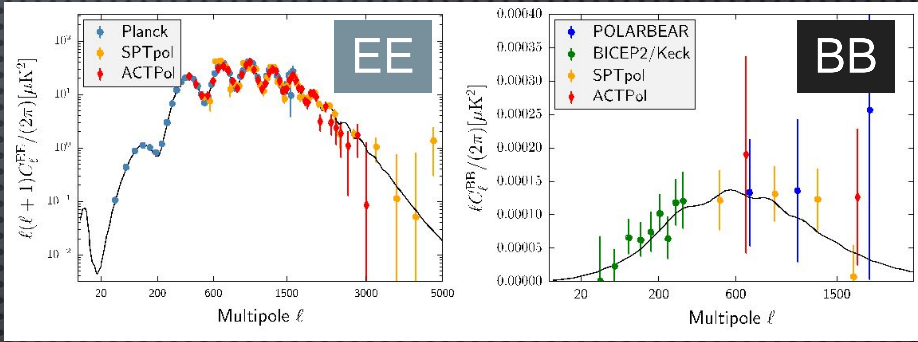
Example from ACT DR6 (Naess et al 2025)



Characteristic plaid patterns in Q,U indicate E-mode dominance in maps

E-Modes and Plaid Patterns in Q & U





2d Fourier Transform equations pick out horizontal and vertical features in an E-only sky.

Flat sky approximations: Zaldarriaga, 2001, PRD 64

$$Q(\boldsymbol{\ell}) = E(\boldsymbol{\ell})\cos(2\phi_\ell) - B(\boldsymbol{\ell})\sin(2\phi_\ell),$$
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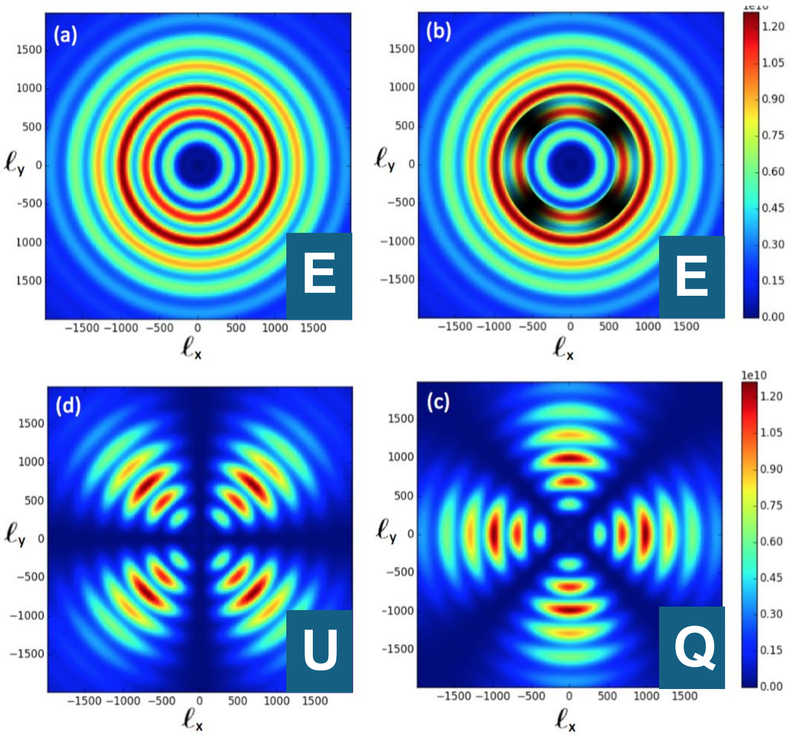
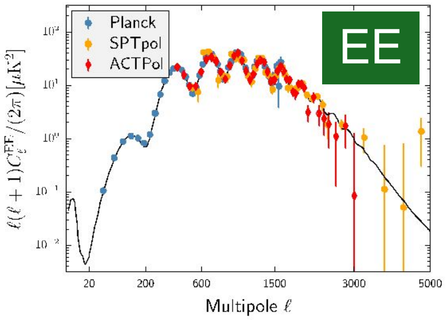
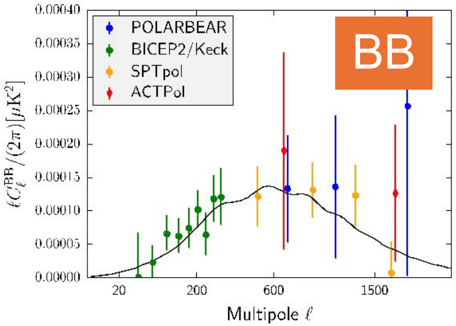
Simulations courtesy T. Louis (from Staggs, Page, Dunkley, 2017.)







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Simulations courtesy T. Louis (from Staggs, Page, Dunkley, 2017.)

OUTLINE:

1) CMB as treasure trove / layer cake / archaeological strata
2) Upcoming new CMB data / instruments in brief
3) Polarization, Parity, Fourier Space, Real Space, Q, U, E, B
4) **Deeper dive on CMB instruments**

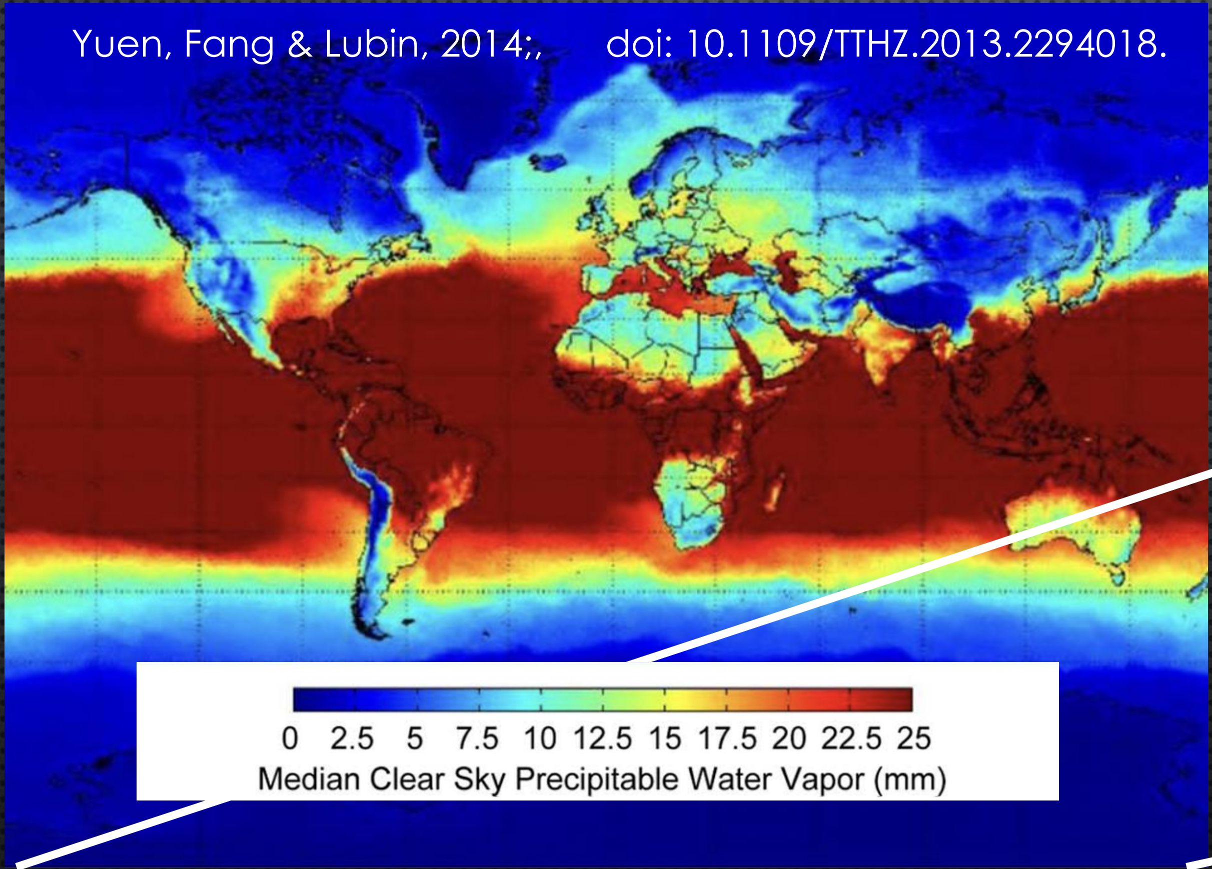
Rationale for New CMB Instruments (including SO)

1)The CMB is packed with fundamental information*
2)Some of that information is encoded subtlely
3)We want it all

Things driving the design of CMB instruments:

- **platform** (**<u>ground</u>**, balloon, satellite)
  - these slides focus on ground-based instruments
- **sensitivity**
  - mandates 100 mK detectors & cold optics – and lots of them
  - modulation is needed (fast scanning, polarization modulators)
- calibration
  - some calibration is built in; some requires external systems
- science driver
  - primordial B-modes: small aperture telescopes are ok
  - high-ell science: large apertures are needed
  - these slides don't address spectral distortion instruments

ATMOSPHERE. ATMOSPHERE. ATMOSPHERE. ATMOSPHERE.

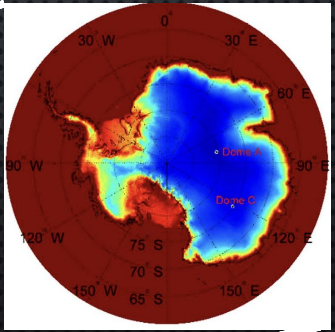


It gets in the way.

(Thus the appeal of balloons & satellites!)

Go dry or go home!

Except for f < 20 GHz



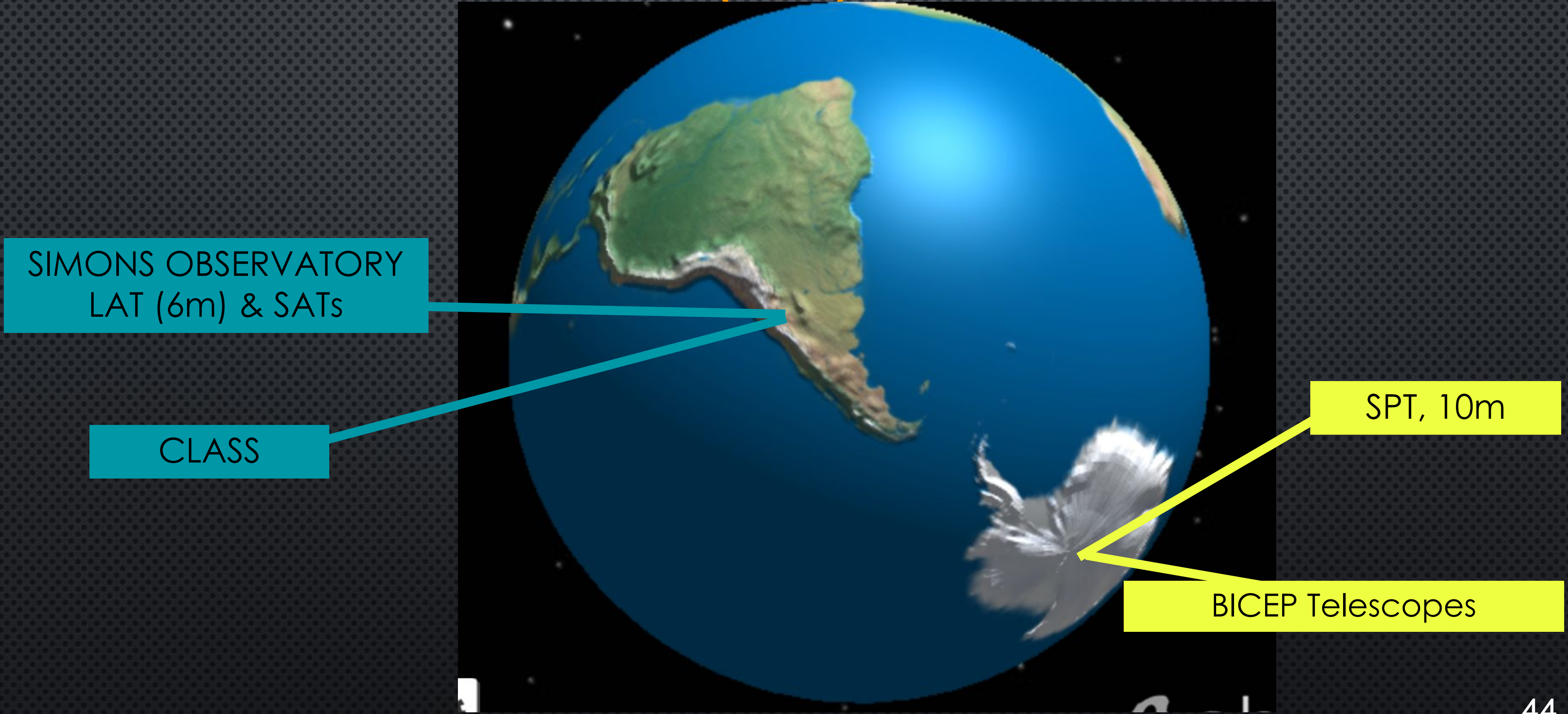
Atacama Desert

Antarctica

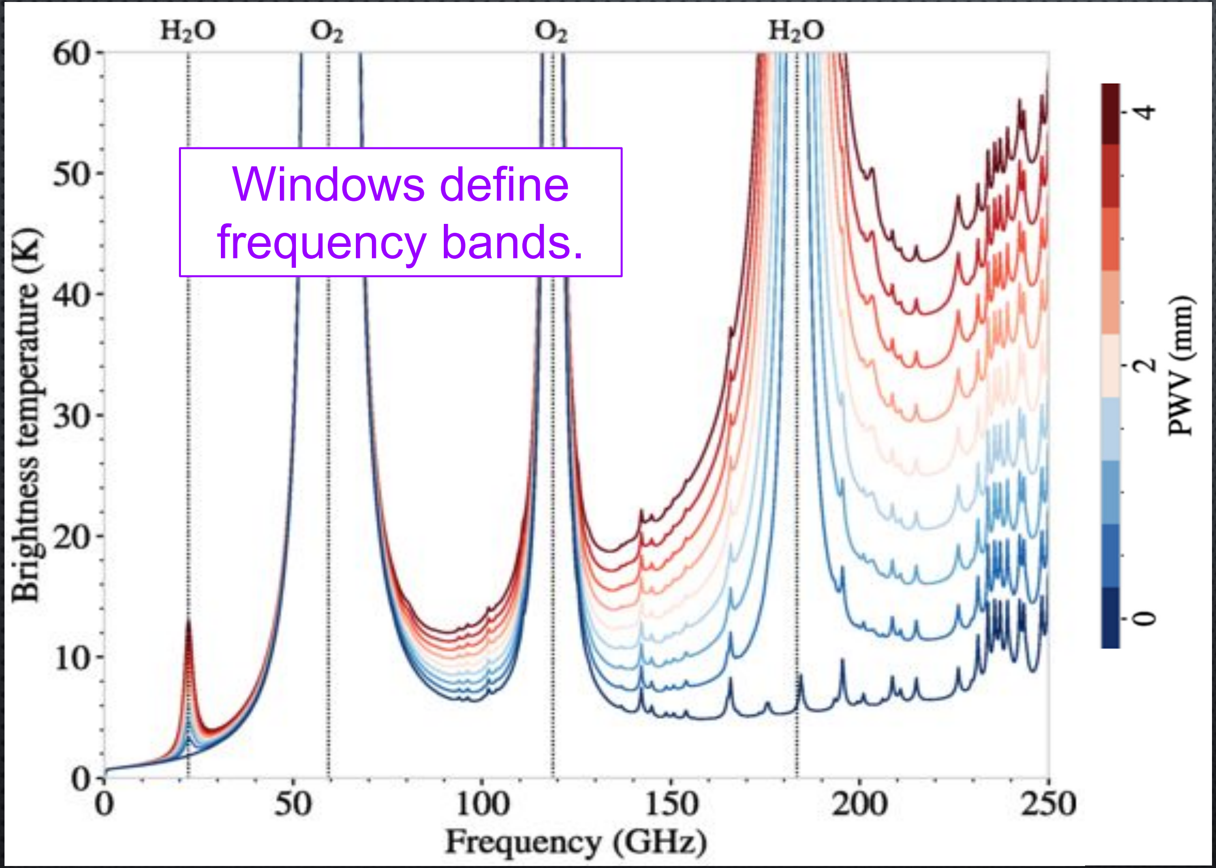
Tibetan Plateau

Greenland

DRY SITES & CURRENT(ISH) INSTRUMENTATION



ATMOSPHERE. ATMOSPHERE. ATMOSPHERE. ATMOSPHERE.



It gets in the way.

And it adds spurious signal.

Extra signal = extra loading.

The signal varies with elevation,

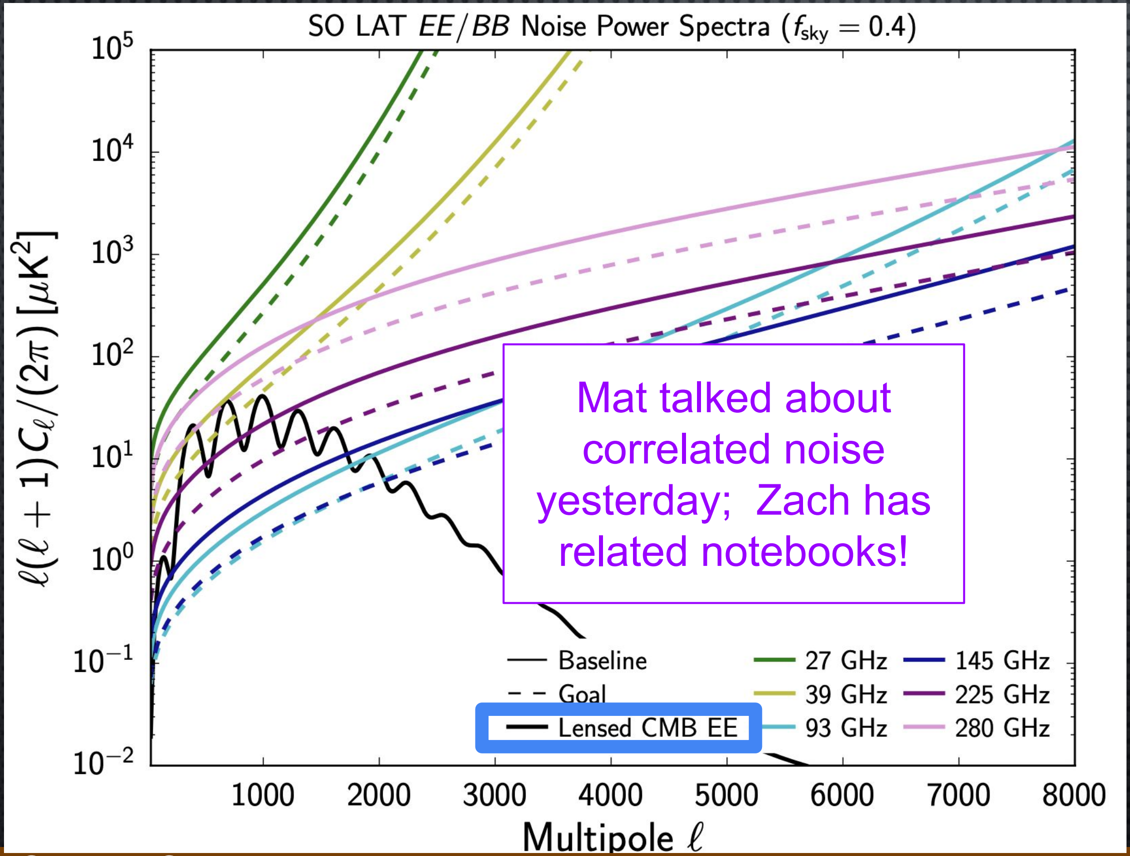
The signal is not constant.

PWV varies with time.

PWV varies with position.

Figure adapted from Morris et al, 2022, PRD 105, 042004; using the am software (S. Paine, 10.5281/ zenodo.5794521)

PWV VARIATIONS LEAD TO EXTRA LOW- L NOISE



Mat talked about correlated noise yesterday; Zach has related notebooks!

EXAMPLE: SO LAT (EE/BB)

Focus on solid dark blue 150 GHz curve

Polarization is less impacted by 1/f (empirically).

These plots show the SO forecasting noise model, based on ACTpol data.

The beam is apparent at high ℓ and the atmosphere at low ℓ.

From SO Collaboration,'The Simons Observatory: Science Goals and Forecasts,' 2019 JCAP02(2019) 056.

SENSITIVITY FRONTIER

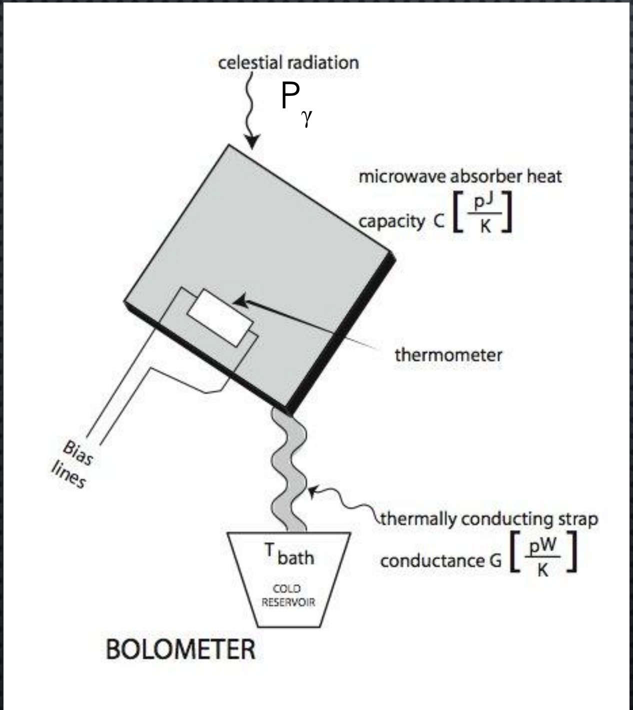
IMPROVING CMB EXPERIMENTS MEANS IMPROVING THE EXPERIMENTAL SENSITIVITY FIRST BY MAKING THE DETECTORS AS SENSITIVE AS POSSIBLE A& THEN BY PUTTING MORE DETECTORS ON THE SKY

GROUND-BASED DETECTORS ARE NOT AS SENSITIVE AS SPACE-BASED ONES DUE TO THE EXTRA SIGNAL FROM THE ATMOSPHERE

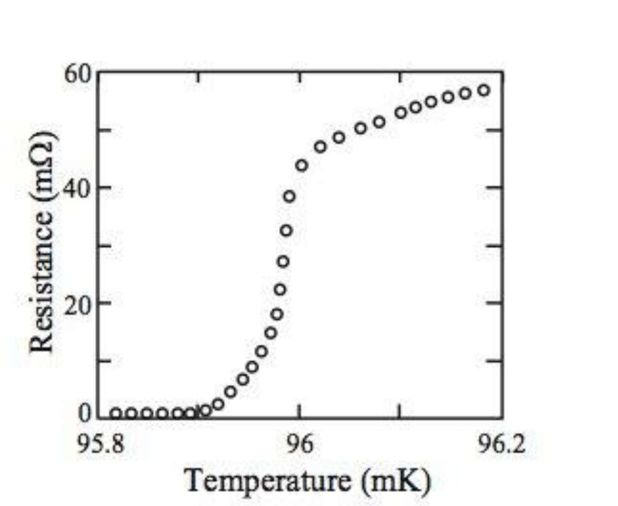
SOMETIMES YOU NEED A BUNCH OF TELESCOPES TO HOUSE THEM ALL

FROM THE GROUND YOU CAN DO THAT.   IT IS CHALLENGING.

TES BOLOMETERS



Use superconductor near T_c as thermometer:



Voltage bias → electrothermal feedback

Total power warming the bolometer is $P_\gamma + P_J$

+

P_γ ↑ means T ↑
T ↑ **means** R ↑
R ↑ means **P_J** ↓

TES Bolometers & Their Multiplexing are Game Changing for Sensitivity

However they bring about some peculiarities;

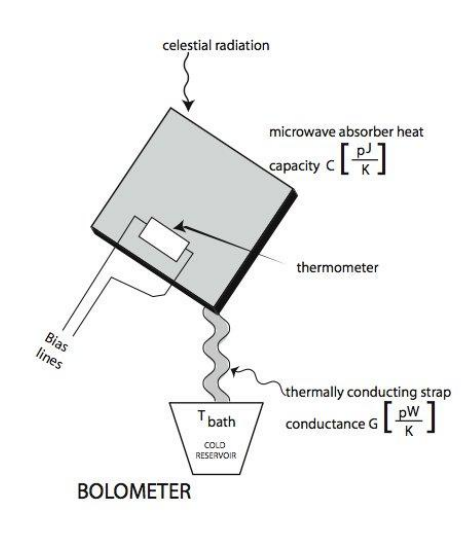
namely, the detectors cease to work if there is too much signal but also cease to work if there is too little signal!

moreover the multiplexed readout uses SQUIDs to measure the tiny current fluctuations; SQUIDs respond quite nonlinearly.

Finally, with 10s of 1000s of detectors, you only understand the “typical behavior” most of the time and rely on cuts to get rid of atypical detectors.

DETECTOR SENSITIVITIES: BOLOMETERS

NEP (units W $Hz^{-1/2}$) and NET (units mK $s^{1/2}$)



$$\mathrm{NEP}_{Dicke} = \sqrt{\frac{p_\gamma^2}{b\,\Delta\nu}}$$

$$\mathrm{NEP}_{shot} = \sqrt{\frac{h\nu\, p_\gamma}{b}}$$

$$\mathrm{NEP}_G = \sqrt{4kT_0^2\, G\, f_\ell}$$

p_γ: the incident photon power
b: conversion factor (0.5) to get units $\mathrm{Hz}^{-1/2}$
$\Delta\nu$: bandwidth
ν: center frequency
f_ℓ: context dependent factor ≤ 1

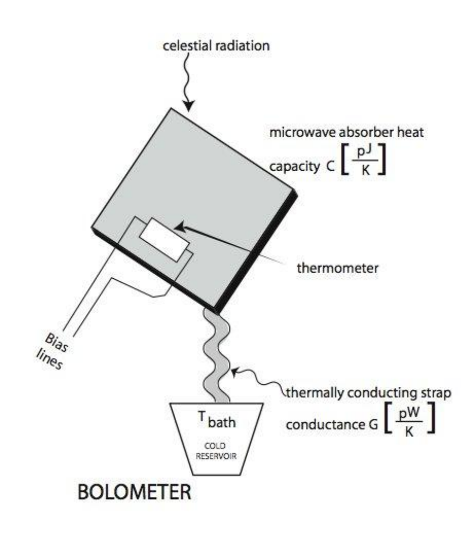
Add these (++) in quadrature to get the total.

Note the NEP_G enters when the input power is transduced into a thermal signature for some kind of thermometer.

Mather, 1982 (Applied Optics)
Zmuidzinas, 2003 (Applied Optics)

DETECTOR SENSITIVITIES: BOLOMETERS

NEP (units W $\mathrm{Hz}^{-1/2}$) and NET (units mK $\mathrm{s}^{1/2}$)



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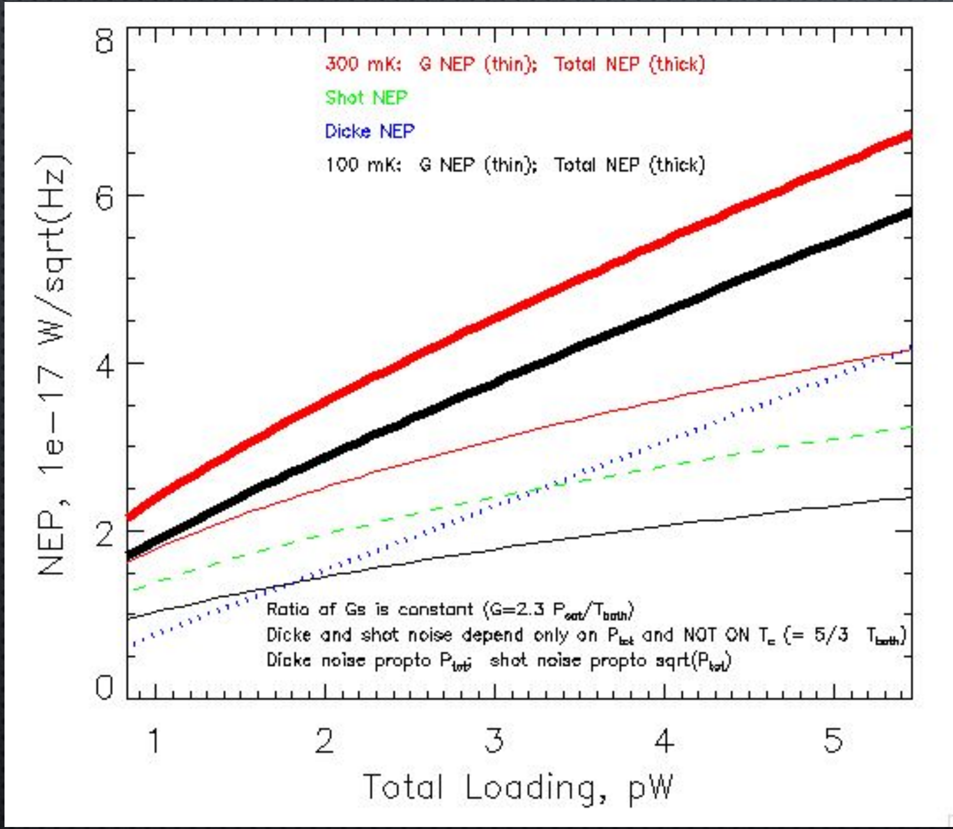
p_γ: the incident photon power
b: conversion factor (0.5) to get units $\mathrm{Hz}^{-1/2}$
$\Delta\nu$: bandwidth
ν: center frequency
f_ℓ: context dependent factor ≤ 1

NEP_{Dicke} and NEP_{shot} both get worse when p_γ increases.
Atmosphere increases p_γ
NEP_G also gets worse, since G increases so $G(T_0 - T_{bath}) > p_\gamma$

Only one thing helps!

DETECTOR SENSITIVITIES: 100 mK & "BACKGROUND LIMITED"



MODERN PULSE-TUBE BACKED DILUTION FRIDGES ARE EASIER TO USE THAN 3He SYSTEMS
(more cooling power – 100-300 uW typical at 100 mK)

We've been running one at 17kft in Chile since 2012.

CHALLENGE OF ATMOSPHERE: SENSITIVITY REQUIRES LOTS OF 100 mK DETECTORS

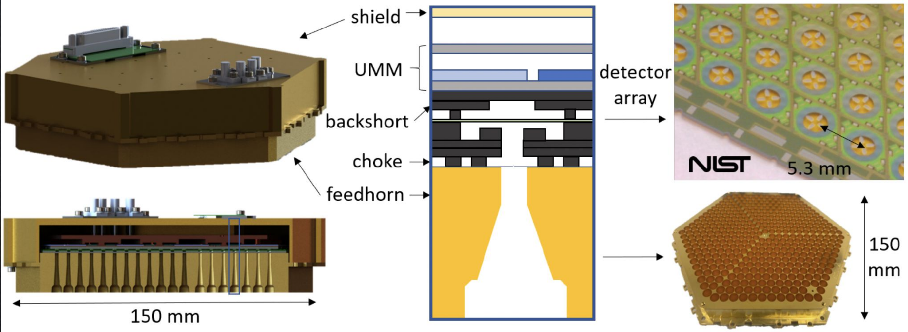
SO, BICEP Array, CMB-S4, Ali-CPT & others:
detector modules based on 150 mm detector wafers

Then, 100,000 detectors ~ 50-100 modules.

Also requires highly multiplexed readout!

PARCHIAL EXAMPLE of SO:

60 modules (compared to 3 modules for ACT DR6)



McCarrick, Healy et al, 2021, for JLTP, arXiv:2112.01458

Each module has:

1 detector wafer
4 optical coupling wafers, aligned & glued
1 multi-layer interface ("routing") wafer
28 multiplexing chips

1 feedhorn array
1 high-precision mounting tray

1 lid
1 magnetic shield
2 PCBs, one with imbedded flex & MDM
4 coax connections
Pogo pins, tripod springs, washers, screws

Each module requires:

Precision gluing of parts (CTE alignments)
~ 12,000 aluminum wire bonds
~ 100 gold wire bods
Pre-assembly screening of most parts
Two stages of testing at 100 mK

MORE DETECTORS → HUGE RECEIVERS
+ HIGH THROUGHPUT PRODUCTION OF OPTICS

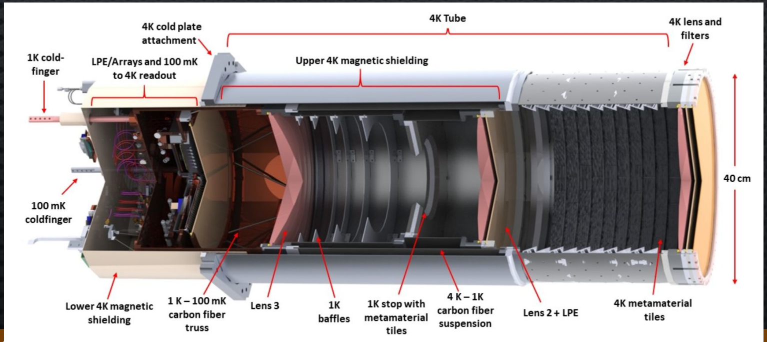
SO LAT Receiver
2.4 m dia

SO LAT holds 13 optics tubes (36-cm) & SO SATs are 6 optics tubes

Each LAT tube has: 3 AR-coated silicon lenses, 1 AR-coated window, 1 AR-coated alumina filter, and free-space filters

Each SO SAT (42 cm stop) has the same elements + an extra AR-coated alumina filter and an AR-coated sapphire HWP

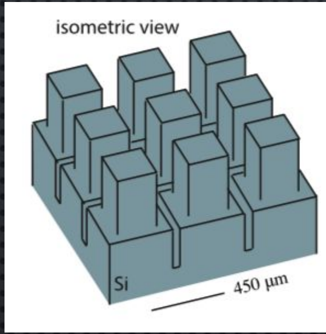
BICEP ARRAY & SPT-3G+ & CCAT also need many AR-coated optical elements!



It could be worse if not for DRs!

Xu et al, 2020; SPIE 11453, doi:10.1117/12.2576151.

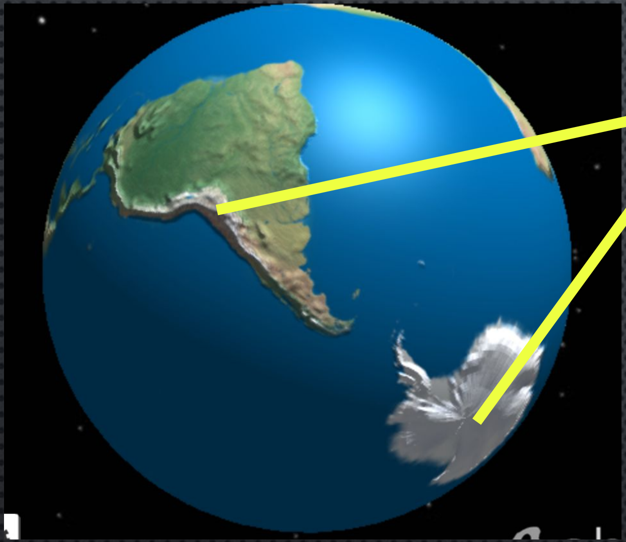
PRODUCTION CHALLENGE: HIGH TRANSMISSION OPTICAL SYSTEMS



Datta et al 2013

ACT, SO, CCAT+: Silicon lenses (n=3.4) with metamaterial AR coating

CHALLENGE OF ATMOSPHERE → SPECIFIC SITES



Great for observations but remote and with complicated stakeholder situations

- Remote →off the grid
  - Power is not cheap
  - Expanding power supply requires civil construction not just $$ to the electric company
- Access for personnel
  - South Pole – (only a) small on-site team has good year-round access to instrument
  - Chile – easy year-round access for entire team, but the instrument site itself is not always accessible

Distributed workforce

- Safety, communications, hiring, shipping

The 'complicated stakeholder situations' at the South Pole have become severe, preventing near-term access for CMB-S4.

ACCESS

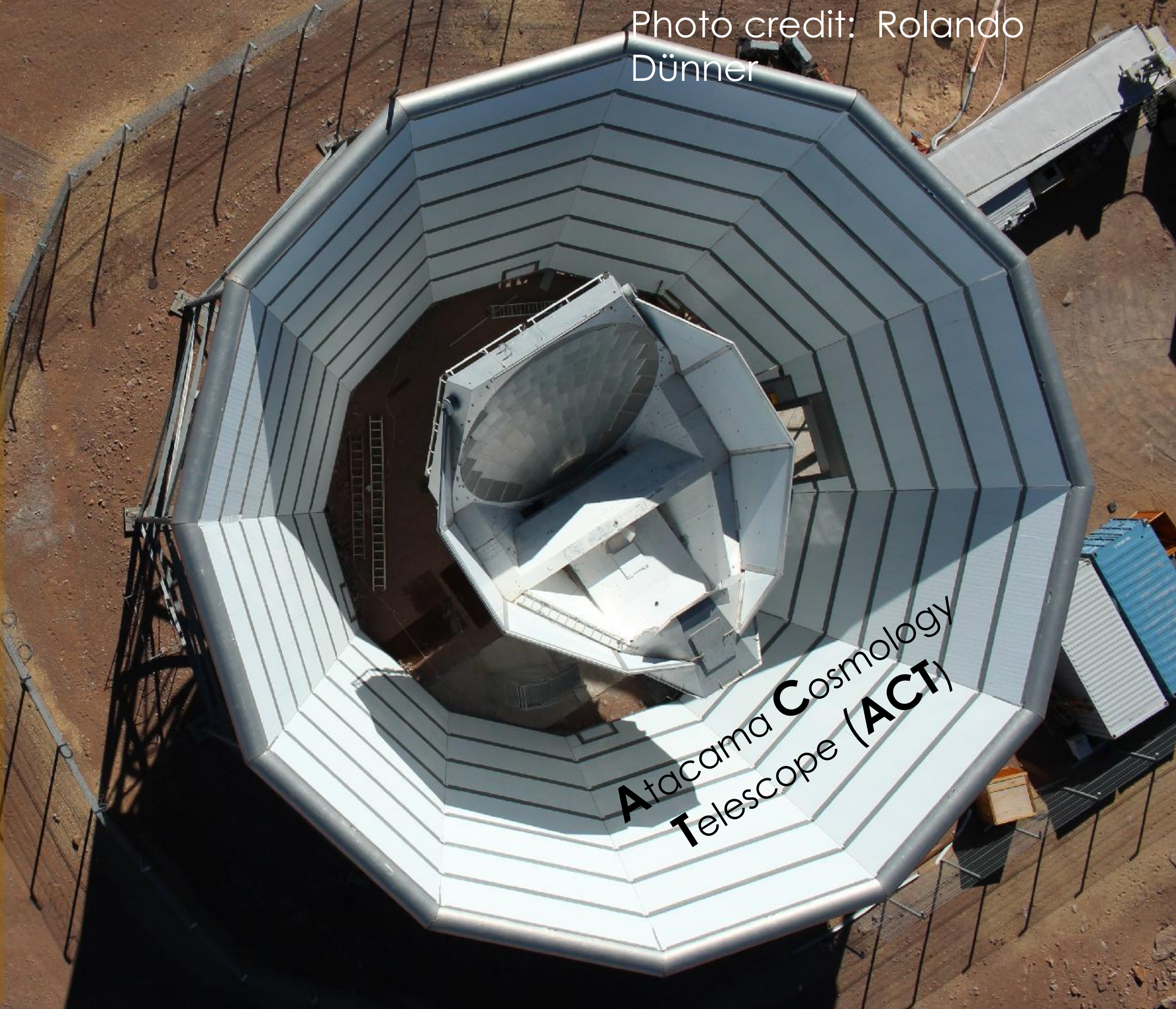
One possible advantage of ground-based instruments may be the ability to use artificial polarized sources to calibrate polarization angles

Another advantage (though necessary) is there are no weight limitations for shielding the telescope.

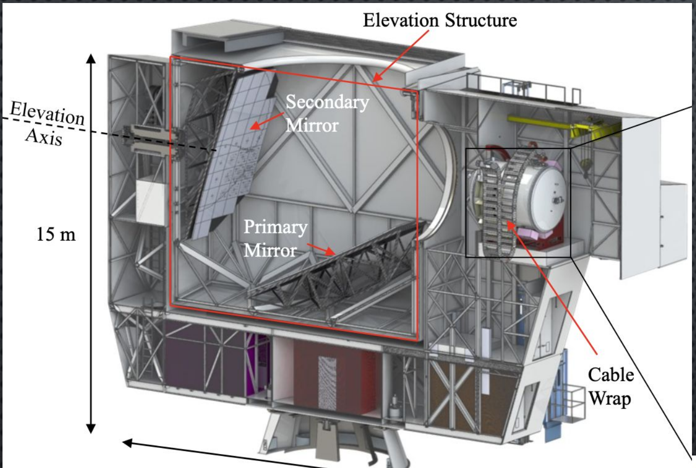
The 6m ACT has a co-moving shield as well as an 11 m fixed shield.

Phased rollout of instrumentation allows modifications to optimize outcomes

Photo credit: Rolando Dünner



EXAMPLES OF DEFENDING AGAINST GROUND PICKUP FOR GROUND-BASED EXPERIMENTS

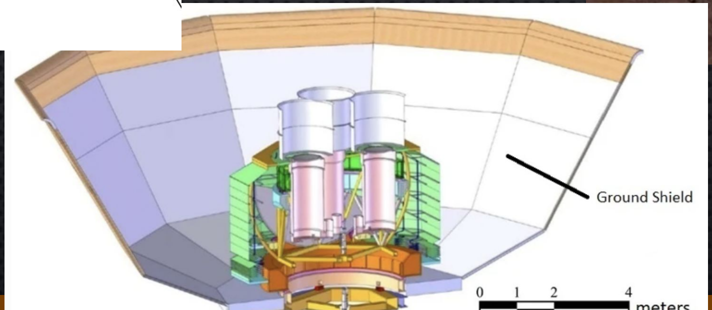


Xu et al, 2020, SPIE Vol. 11453, 1145315,

SO Large Aperture Telescope (LAT); same design for CCAT & CMB-S4

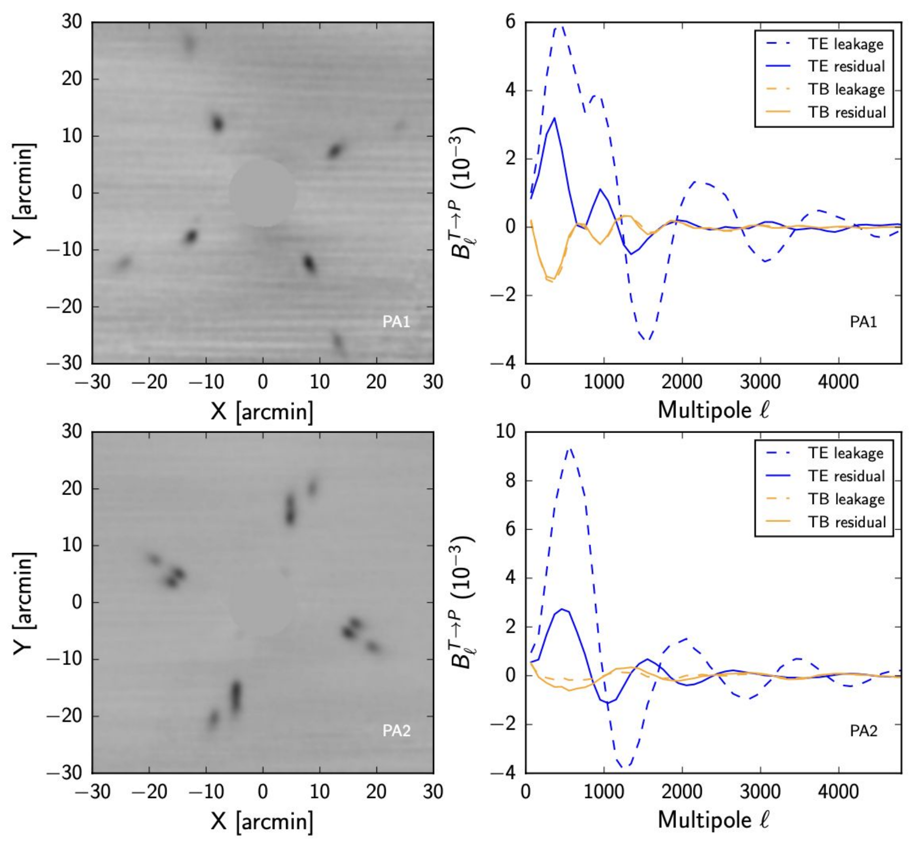
BICEP ARRAY in its large ground screen

SO Small Aperture Telescopes inside their large ground screens

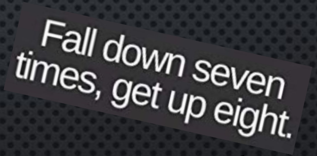


Schillaci et al, 2020, JLTP 199, 976,

MULTIPLE OPTICAL ELEMENTS IN CRYOGENIC OPTICS TUBES CAN LEAD TO SURPRISES



Polarized sidelobe structure in the ACTPol receiver



PROGRESS DESPITE CHALLENGES!

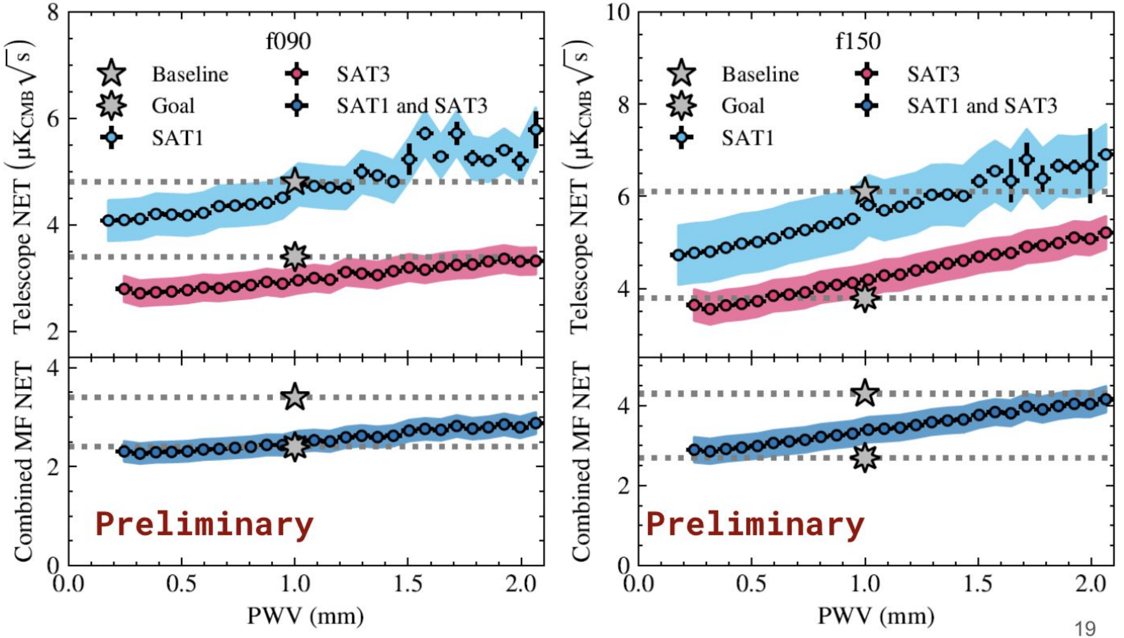
Example from a talk last week at SPIE by Katie Harrington showing the SO SAT performance exceeds requirements.

Telescope NETs

Shaded regions: calibration uncertainties at the end of commissioning

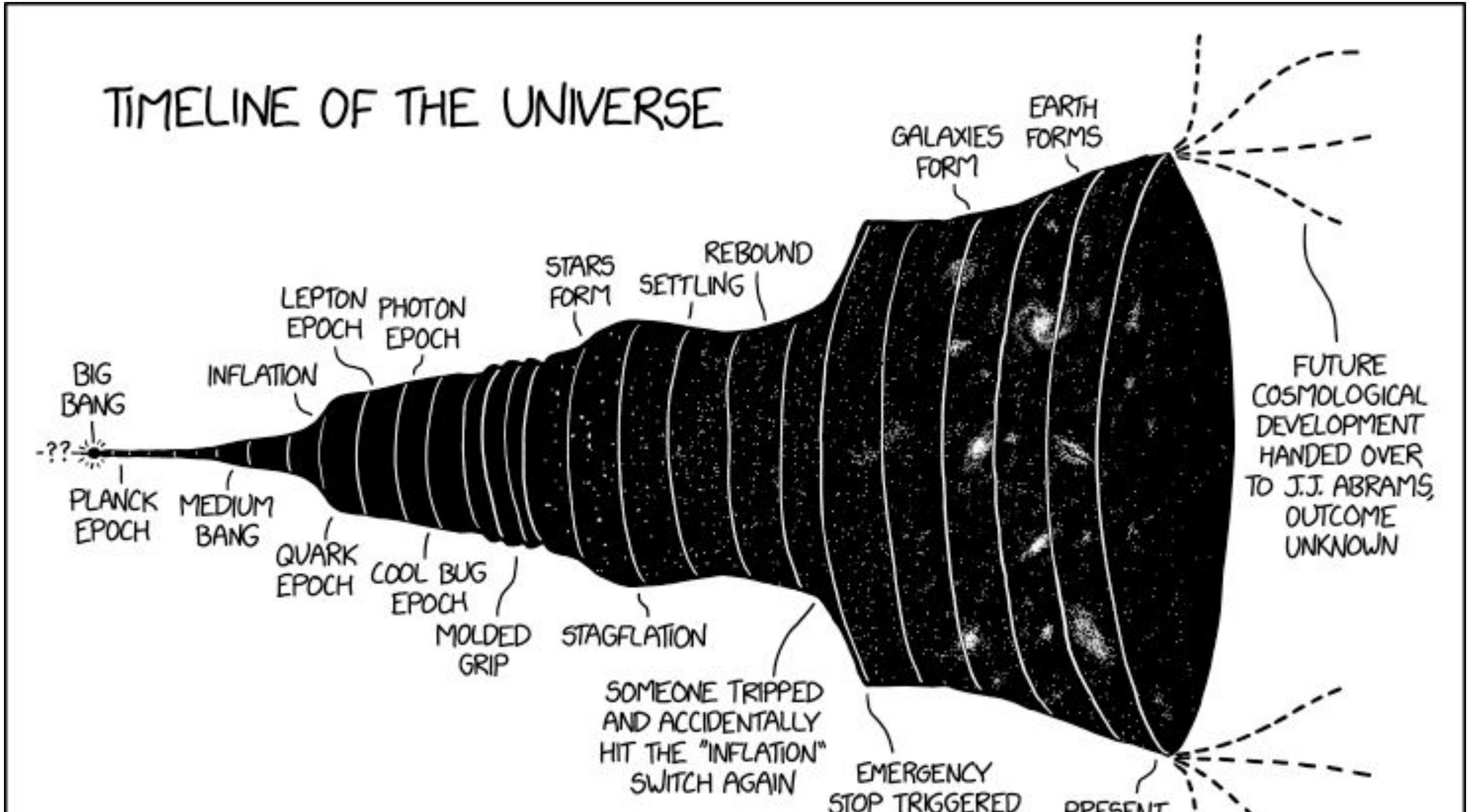
All telescopes performing better than baseline! Commissioning goal achieved!

Combined 90 GHz sensitivity is near goal.



19

END SLIDE


TIMELINE OF THE UNIVERSE
-??-
BIG BANG
PLANCK EPOCH
MEDIUM BANG
INFLATION
QUARK EPOCH
LEPTON EPOCH
PHOTON EPOCH
COOL BUG EPOCH
MOLDED GRIP
STARS FORM
STAGFLATION
SETTLING
REBOUND
SOMEONE TRIPPED AND ACCIDENTALLY HIT THE "INFLATION" SWITCH AGAIN
EMERGENCY STOP TRIGGERED
GALAXIES FORM
EARTH FORMS
PRESENT DAY
FUTURE COSMOLOGICAL DEVELOPMENT HANDED OVER TO J.J. ABRAMS, OUTCOME UNKNOWN

[kxcd 2240]